Introducción al modelo OMOP-CDM

Alberto Labarga – Febrero 2026

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OMOP-CDM
El curso

El curso tiene como objetivo proporcionar una comprensión profunda del modelo de datos **OMOP-CDM** de la OHDSI, incluida su **estructura**, tablas clave y **vocabularios**.

Abordaremos también como **transformar** datos al modelo y explorar y analizar la **calidad** de los datos transformados.

Febrero
2026

L	M	X	J	V	S	D
						1
2	3	4	5	6	7	8
9	**10**	11	**12**	13	14	15
16	17	18	**19**	20	21	22
23	**24**	25	**26**	27	28	1
2	**3**					

Objetivos del curso

01 Introducción al modelo OMOP-CDM, a sus tablas y sus vocabularios

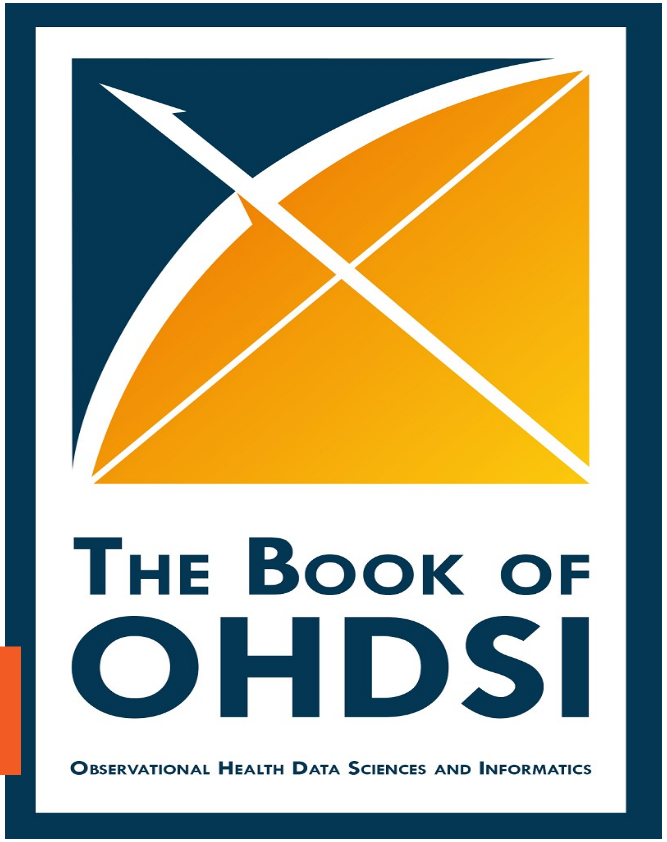
02 Transformación de datos al modelo OMOP-CDM utilizando la suite de herramientas de la OHDSI

03 Análisis de calidad y explotación de una base de datos OMOP-CDM

Recursos de aprendizaje

Sitio web: https://github.com/alabarga/curso-omop-cdm

Libro https://ohdsi.github.io/TheBookOfOhdsi/



OMOP-CDM

VOCABULARIOS

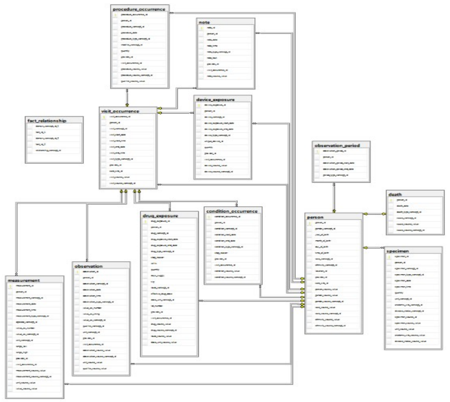

SJD
Sant Joan de Déu
Barcelona · Hospital
INGEMM
idis
IMIBIC
IBSAL

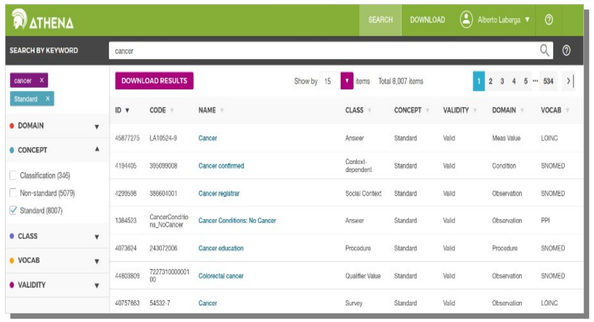
OMOP-CDM

Common Data Model

OMOP-CDM pretende servir como modelo de datos común para poder comprender los datos clínicos sin importar su origen. Este es un estándar de **licencia abierta** cuyo objetivo es estandarizar los datos y estructurarlos para permitir análisis y observaciones eficientes cuyos resultados sean fiables.

Propone **interoperabilidad sintáctica** (modelo de datos) e **interoperabilidad semántica** (vocabularios)

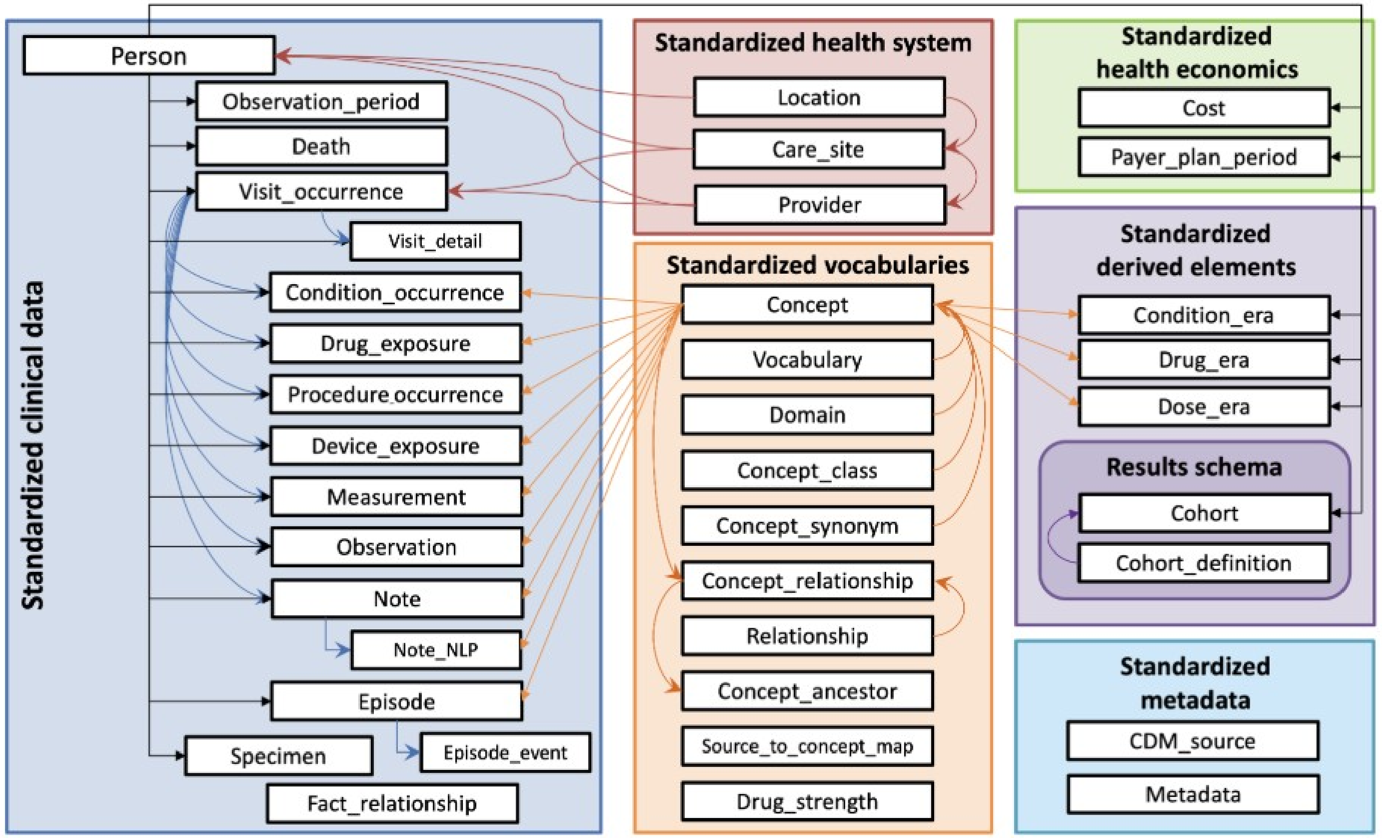




OMOP Common Data Model | Background | Conventions | CDM Versions | CDM Additions | ? How to | Support

The current CDM version is CDM v5.4, depicted below. This CDM version was developed over the course of a year by considering requests that were sent via our issues page. The list of proposed changes was then shared with the community in multiple ways: through discussions at the weekly OHDSI Community calls, discussions with the OHDSI Steering Committee, and discussions with all potentially affected workgroups. The final changes were then delivered to the Community through a new R package designed to dynamically generate the DDLs and documentation for all supported SQL dialects. Looking for an entity-relationshop diagram? Click here!

- Link to DDLs for CDM v5.4
- Link to ReadMe for instructions on how to use the R package



CommonDataModel / inst / ddl / 5.4 /

clairblacketer Removing codespell and older ddl folder for CRAN submission

Name
..
bigquery
duckdb
hive
impala
netezza
oracle
pdw
postgresql
redshift
snowflake
spark
sql_server
sqlite
sqlite_extended
synapse

https://ohdsi.github.io/CommonDataModel/

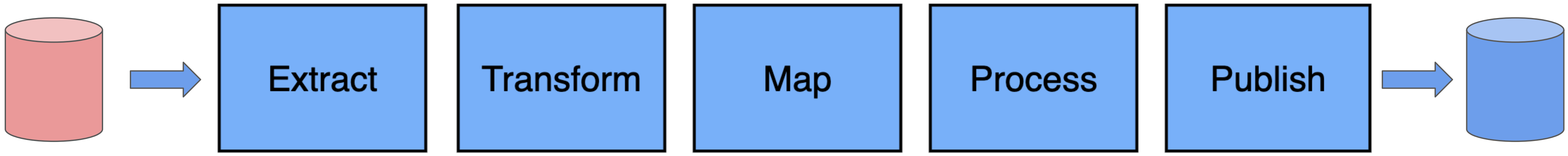
ATHENA | SEARCH | DOWNLOAD | Alberto Labarga

Show all | SHOW HISTORY | DOWNLOAD VOCABULARIES

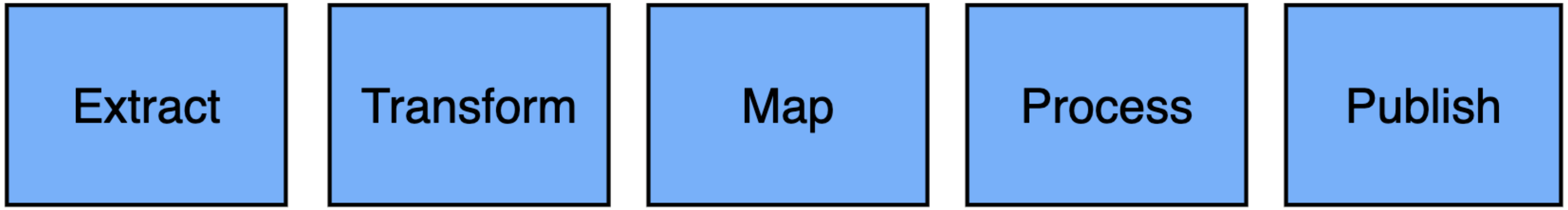
ID (CDM V4.5)	CODE (CDM V5)	NAME	REQUIRED	LATEST UPDATE
1	SNOMED	Systematic Nomenclature of Medicine - Clinical Terms (IHTSDO)		01-Mar-2025
2	ICD9CM	International Classification of Diseases, Ninth Revision, Clinical Modification, Volume 1 and 2 (NCHS)		01-Oct-2014
3	ICD9Proc	International Classification of Diseases, Ninth Revision, Clinical Modification, Volume 3 (NCHS)		01-Oct-2014
4	CPT4	Current Procedural Terminology version 4 (AMA)	EULA required	05-May-2025
5	HCPCS	Healthcare Common Procedure Coding System (CMS)		01-Jul-2025
6	LOINC	Logical Observation Identifiers Names and Codes (Regenstrief Institute)		26-Feb-2025
7	NDFRT	National Drug File - Reference Terminology (VA)		06-Aug-2018
8	RxNorm	RxNorm (NLM)		02-Jun-2025
9	NDC	National Drug Code (FDA and manufacturers)		17-Aug-2025

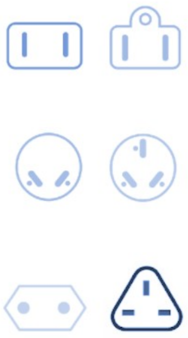
https://athena.ohdsi.org

TRANSFORMACIÓN
AL MODELO OMOP-CDM



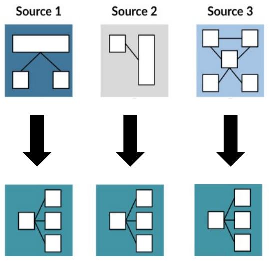
TRANSFORMACIÓN
AL MODELO OMOP-CDM





TRANSFORMACIÓN
AL MODELO OMOP-CDM

Extract | Transform | Map | Process | Publish



TRANSFORMACIÓN
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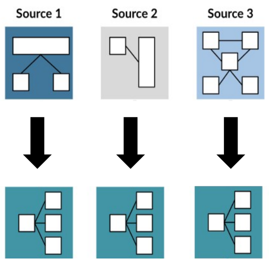
Extract | Transform | Map | Process | Publish

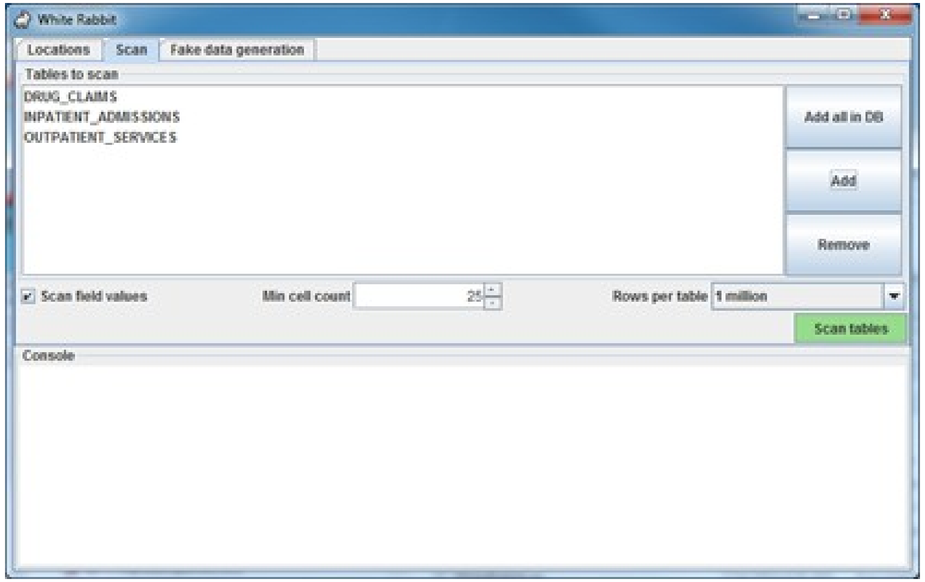


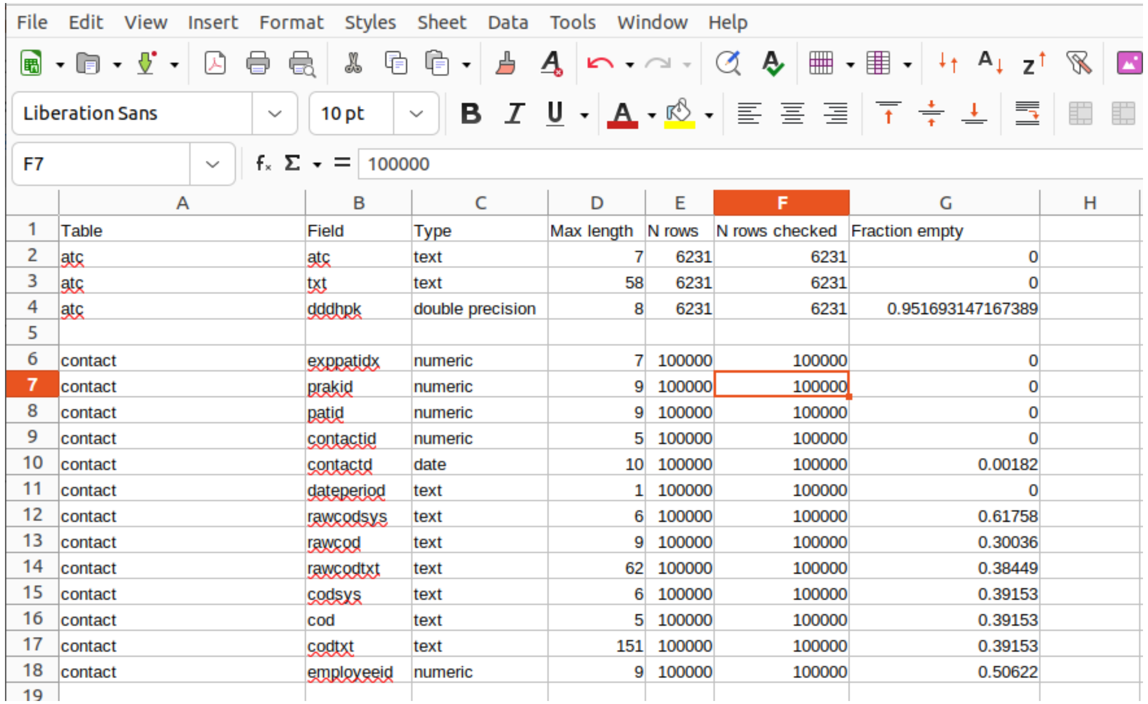
WHITE RABBIT

RABBIT IN A HAT

WhiteRabbit

DATA PROFILING



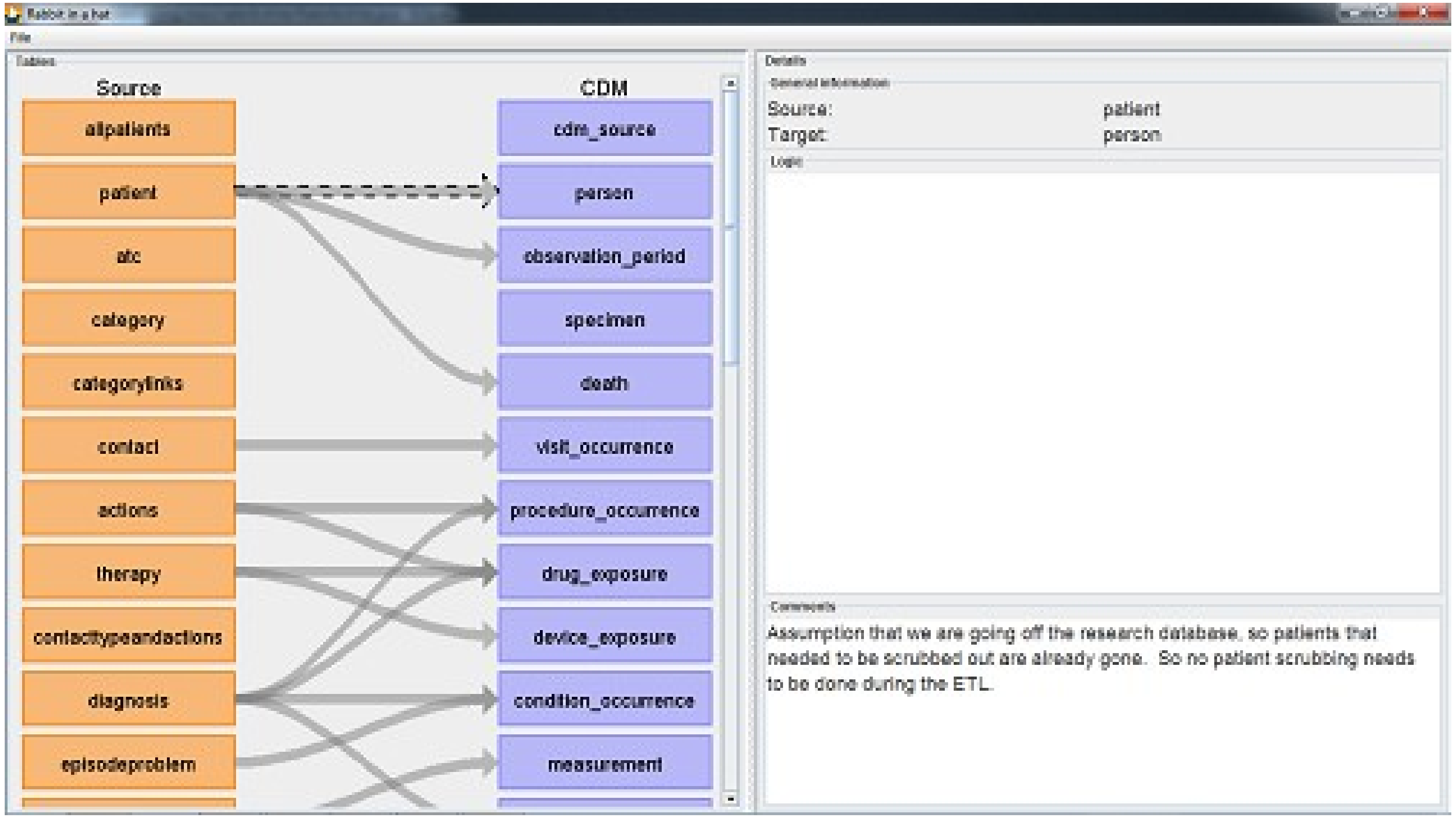


	A	B	C	D	E	F	G
1	Table	Field	Type	Max length	N rows	N rows checked	Fraction empty
2	atc	atc	text	7	6231	6231	0
3	atc	txt	text	58	6231	6231	0
4	atc	dddhpk	double precision	8	6231	6231	0.951693147167389
5							
6	contact	exppatidx	numeric	7	100000	100000	0
7	contact	prakid	numeric	9	100000	100000	0
8	contact	patid	numeric	9	100000	100000	0
9	contact	contactid	numeric	5	100000	100000	0
10	contact	contactd	date	10	100000	100000	0.00182
11	contact	dateperiod	text	1	100000	100000	0
12	contact	rawcodsys	text	6	100000	100000	0.61758
13	contact	rawcod	text	9	100000	100000	0.30036
14	contact	rawcodtxt	text	62	100000	100000	0.38449
15	contact	codsys	text	6	100000	100000	0.39153
16	contact	cod	text	5	100000	100000	0.39153
17	contact	codtxt	text	151	100000	100000	0.39153
18	contact	employeeid	numeric	9	100000	100000	0.50622
19							

https://github.com/OHDSI/WhiteRabbit

Rabbit -in-a-Hat

DATA MAPPING



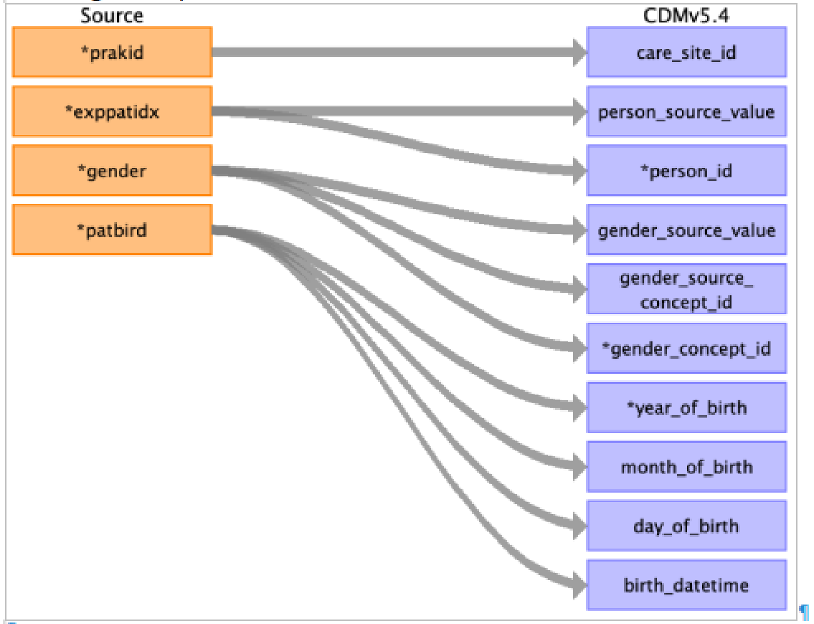
https://github.com/OHDSI/WhiteRabbit

Rabbit -in-a-Hat

DATA MAPPING

Table name: person

Reading from patient

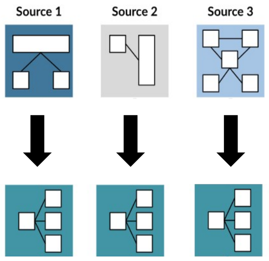


Destination Field	Source Field	Logic	Comment
care_site_id	prakid	Copy source value	
person_source_value	exppatidx	Copy source value	In many situations, we would generate a surrogate key for pseudonimization and get person_id through this field in Clinical Tables
person_id	exppatidx	Copy source value	In many situations, we would generate a surrogate key for

https://github.com/OHDSI/WhiteRabbit

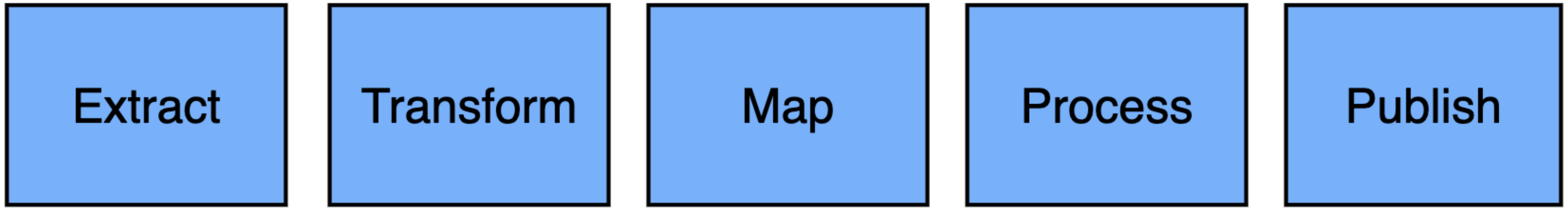
TRANSFORMACIÓN
AL MODELO OMOP-CDM

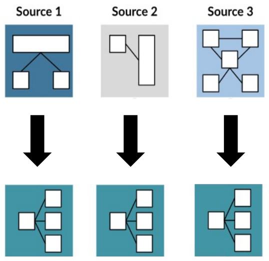
Extract | Transform | Map | Process | Publish



dbt

TRANSFORMACIÓN
AL MODELO OMOP-CDM






```sql
select
    {{ create_id_from_str('"Id"::text')}} AS person_id,
    {{ gender_concept_id ('"GENDER"') }} AS gender_concept_id,
    date_part('year', "BIRTHDATE"::DATE)::INT AS year_of_birth,
    date_part('month', "BIRTHDATE"::DATE)::INT AS month_of_birth,
    date_part('day', "BIRTHDATE"::DATE)::INT AS day_of_birth,
    "BIRTHDATE"::TIMESTAMP AS birth_datetime,
    {{ race_concept_id('"RACE"') }}  AS race_concept_id,
    {{ ethnicity_concept_id('"ETHNICITY"') }} AS ethnicity_concept_id,
    NULL::INT AS location_id,
    NULL::INT AS provider_id,
    NULL::INT AS care_site_id,
    "Id"::VARCHAR(50) AS person_source_value,
    "GENDER"::VARCHAR(50) AS gender_source_value,
    0 AS gender_source_concept_id,
    "RACE"::VARCHAR(50) AS race_source_value,
    0 AS race_source_concept_id,
    "ETHNICITY"::VARCHAR(50) AS ethnicity_source_value,
    0 AS ethnicity_source_concept_id
from patients
where "BIRTHDATE" is not null -- Don't load patients who do not have birthdate and sex (change variable
names if necessary)
      and "GENDER" is not null
  return go(f, seed, [])
}
```

TRANSFORMACIÓN
AL MODELO OMOP-CDM

Extract | Transform | Map | Process | Publish

CONCEPT_ID	313217	For use in CDM
CONCEPT_NAME	Atrial fibrillation	English description
DOMAIN_ID	Condition	Domain
VOCABULARY_ID	SNOMED	Vocabulary
CONCEPT_CLASS_ID	Clinical Finding	Class in SNOMED
STANDARD_CONCEPT	S	Concept in data
CONCEPT_CODE	49436004	Code in SNOMED
VALID_START_DATE	01-Jan-1970	Valid during time interval
VALID_END_DATE	31-Dec-2099	
INVALID_REASON		

TRANSFORMACIÓN
AL MODELO OMOP-CDM

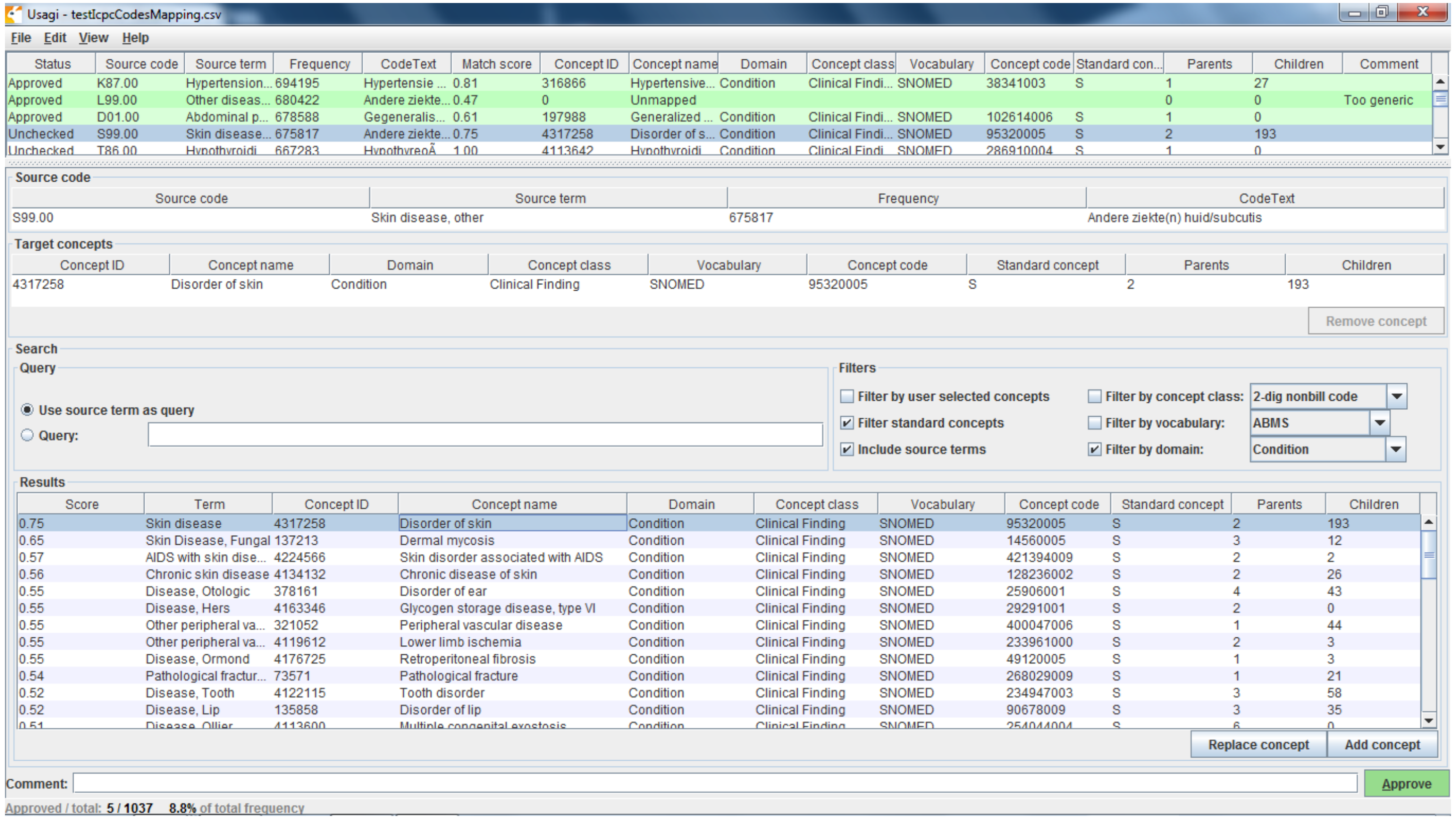
Extract | Transform | Map | Process | Publish

USAGI

CONCEPT_ID	313217	For use in CDM
CONCEPT_NAME	Atrial fibrillation	English description
DOMAIN_ID	Condition	Domain
VOCABULARY_ID	SNOMED	Vocabulary
CONCEPT_CLASS_ID	Clinical Finding	Class in SNOMED
STANDARD_CONCEPT	S	Concept in data
CONCEPT_CODE	49436004	Code in SNOMED
VALID_START_DATE	01-Jan-1970	Valid during time interval
VALID_END_DATE	31-Dec-2099	
INVALID_REASON		

Usagi

CONCEPT MAPPING



https://github.com/OHDSI/Usagi

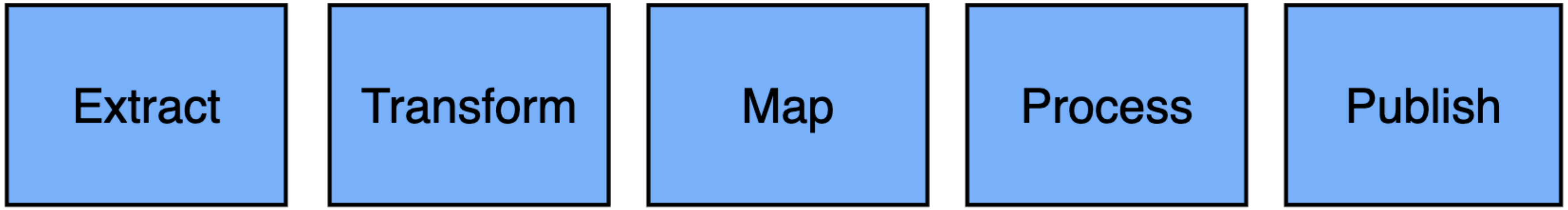
TRANSFORMACIÓN
AL MODELO OMOP-CDM

Extract | Transform | Map | Process | Publish

El paciente acude con su madre Pepita **PER** , con la que vive en Calle Córcega 23 **LOC** , refiriendo dolor abdominal.

El paciente acude con su madre Lucía **PER** , con la que vive en Calle Londres **LOC** , refiriendo dolor abdominal.

TRANSFORMACIÓN
AL MODELO OMOP-CDM



Acude con dolor abdominal derecho de 2 días de duración. Pauta de vacunación completa.

Finding *dolor abdominal.*

Spatial Concept *derecho.*

Temporal Concept *2 días, duración.*

Therapeutic or Preventive Procedure *vacunación.*

Qualitative Concept *completa.*

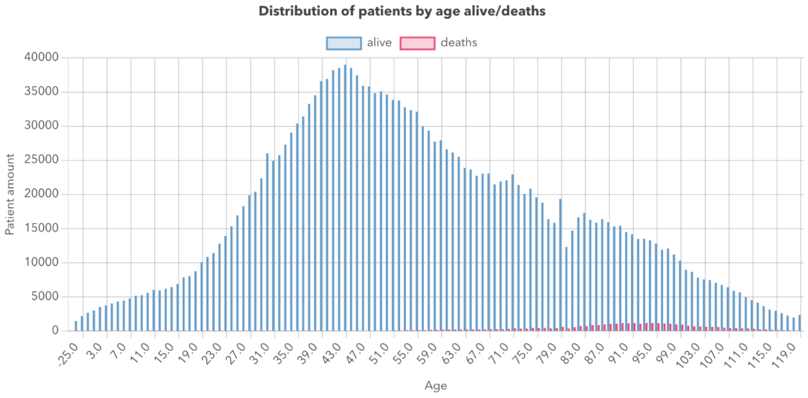
PROCESADO
DEL LENGUAJE NATURAL

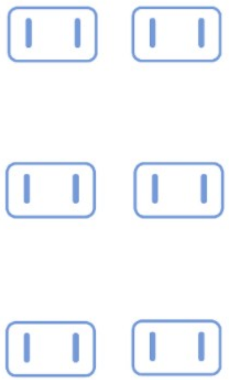
Estadísticas generales de los resultados obtenidos con el procesado de notas.
- Number of patients with notes: 934.000
- Number of notes: 46.430.000
- Average length: 700
- Average number of entities detected per note: 56

Domain	Events in structure data	Events retrieved by NLP
Condition	2.562.120	537.854.624
Procedure	800.717	205.344.592
Observation	335.659	1.621.526.706
Drug	5.002.550	132.818.091
Measurement	185.752.461	158.167.899
Total	194.4M	2.650M

TRANSFORMACIÓN
AL MODELO OMOP-CDM

Extract | Transform | Map | Process | Publish

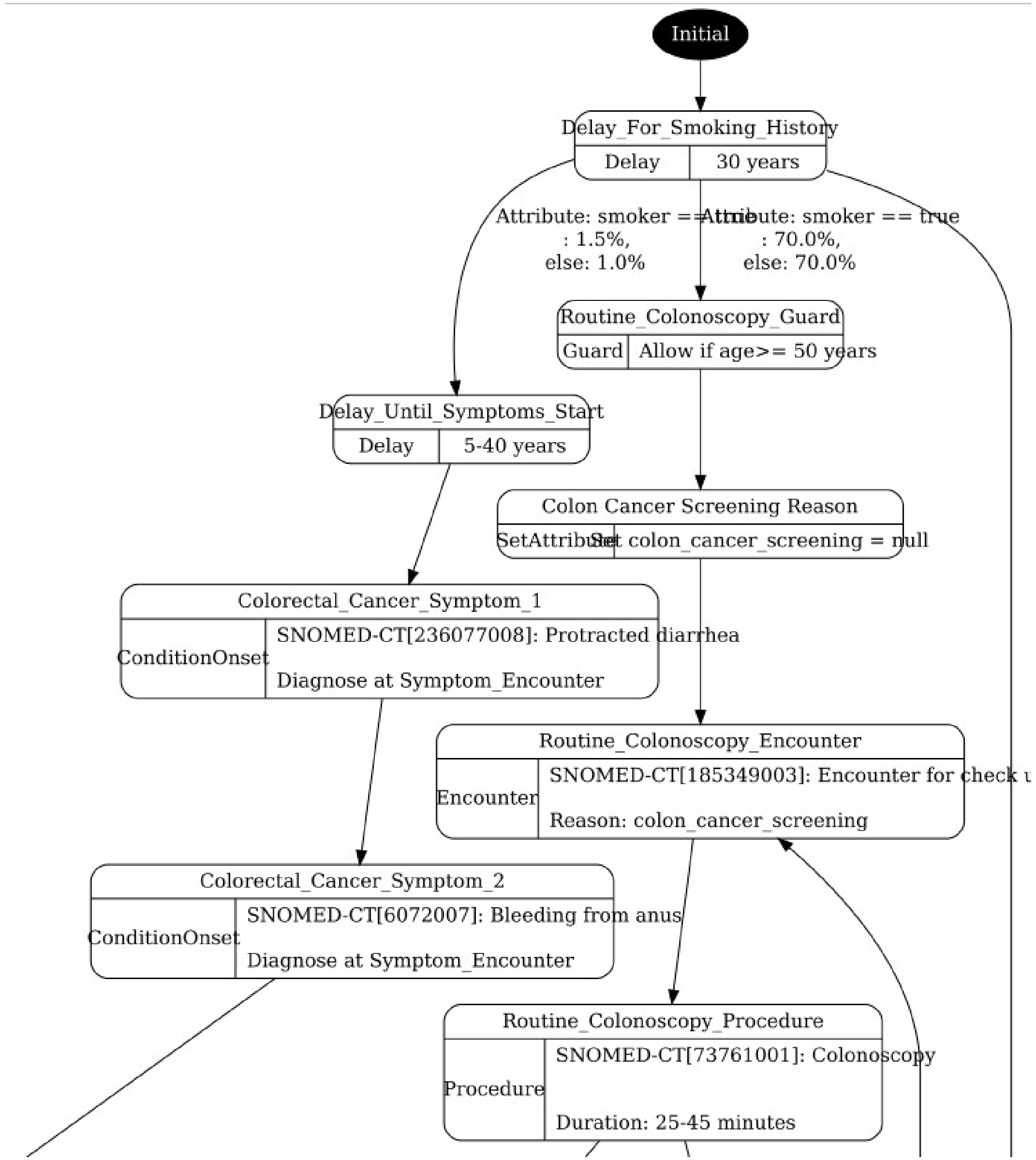




Ejercicio

Transformación a OMOP-CDM

DATOS SINTÉTICOS



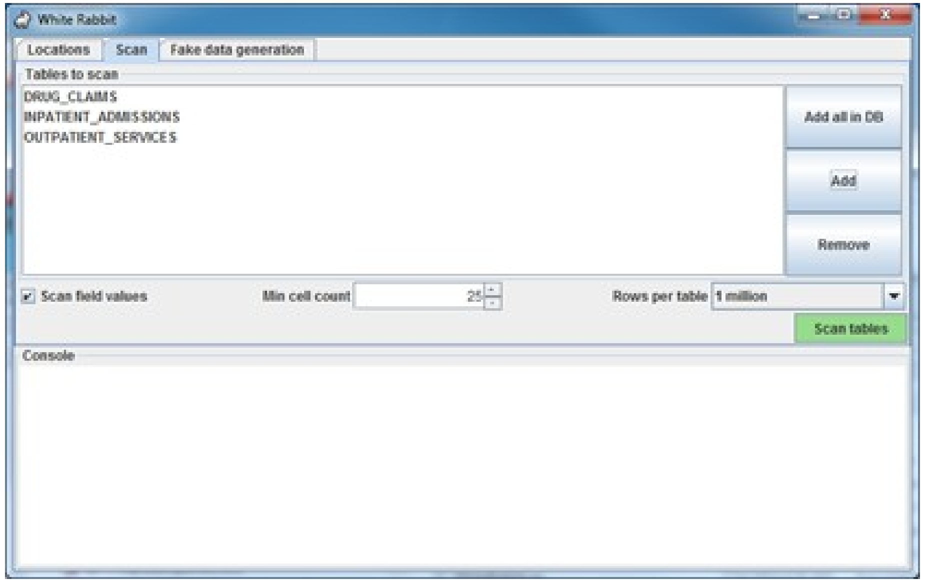
https://synthea.mitre.org/

DATOS SINTÉTICOS

File	Description
allergies.csv	Patient allergy data.
careplans.csv	Patient care plan data, including goals.
claims.csv	Patient claim data.
claims_transactions.csv	Transactions per line item per claim
conditions.csv	Patient conditions or diagnoses.
devices.csv	Patient-affixed permanent and semi-permanent devices.
encounters.csv	Patient encounter data.
imaging_studies.csv	Patient imaging metadata.
immunizations.csv	Patient immunization data.
medications.csv	Patient medication data.
observations.csv	Patient observations including vital signs and lab reports.
organizations.csv	Provider organizations including hospitals.
patients.csv	Patient demographic data.
payer_transitions.csv	Payer Transition data (i.e. changes in health insurance).
payers.csv	Payer organization data.

WhiteRabbit

DATA PROFILING



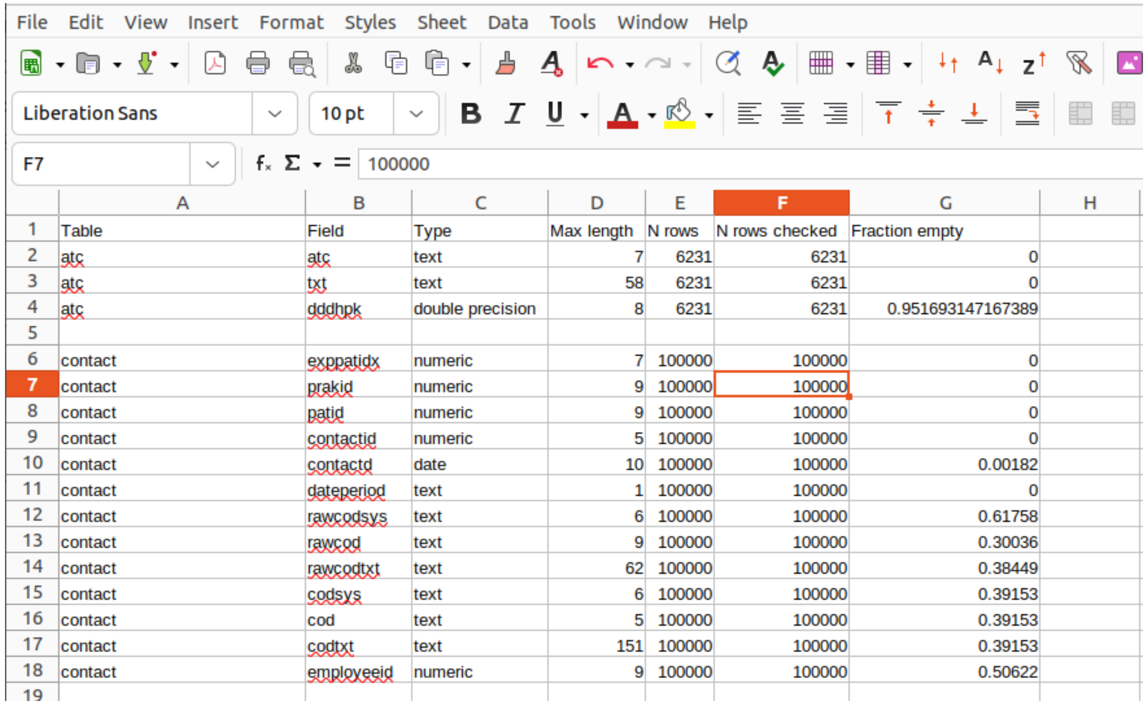


Table	Field	Type	Max length	N rows	N rows checked	Fraction empty
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atc	txt	text	58	6231	6231	0
atc	dddhpk	double precision	8	6231	6231	0.951693147167389
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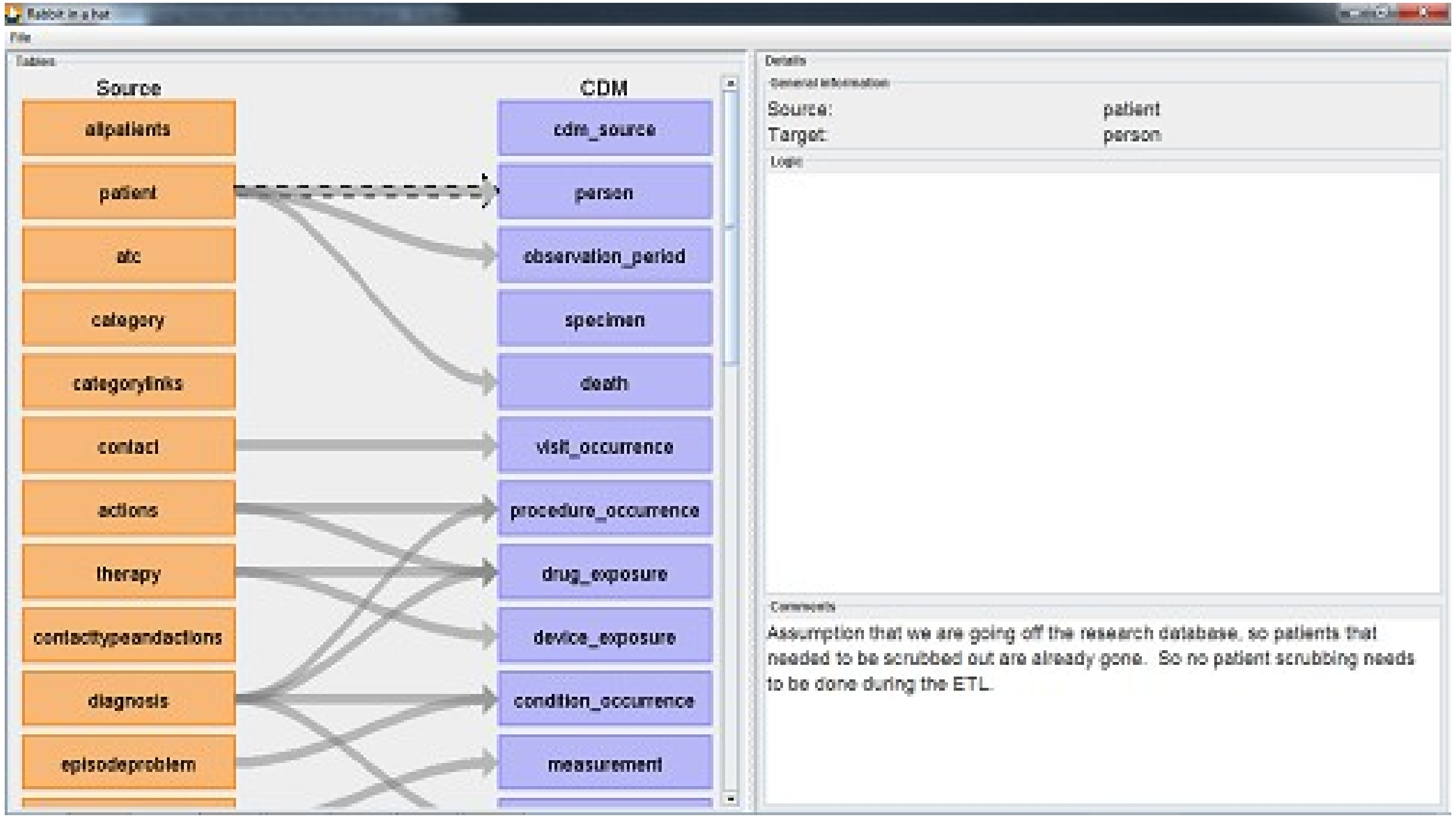
contact	prakid	numeric	9	100000	100000	0
contact	patid	numeric	9	100000	100000	0
contact	contactid	numeric	5	100000	100000	0
contact	contactd	date	10	100000	100000	0.00182
contact	dateperiod	text	1	100000	100000	0
contact	rawcodsys	text	6	100000	100000	0.61758
contact	rawcod	text	9	100000	100000	0.30036
contact	rawcodtxt	text	62	100000	100000	0.38449
contact	codsys	text	6	100000	100000	0.39153
contact	cod	text	5	100000	100000	0.39153
contact	codtxt	text	151	100000	100000	0.39153
contact	employeeid	numeric	9	100000	100000	0.50622

https://github.com/OHDSI/WhiteRabbit

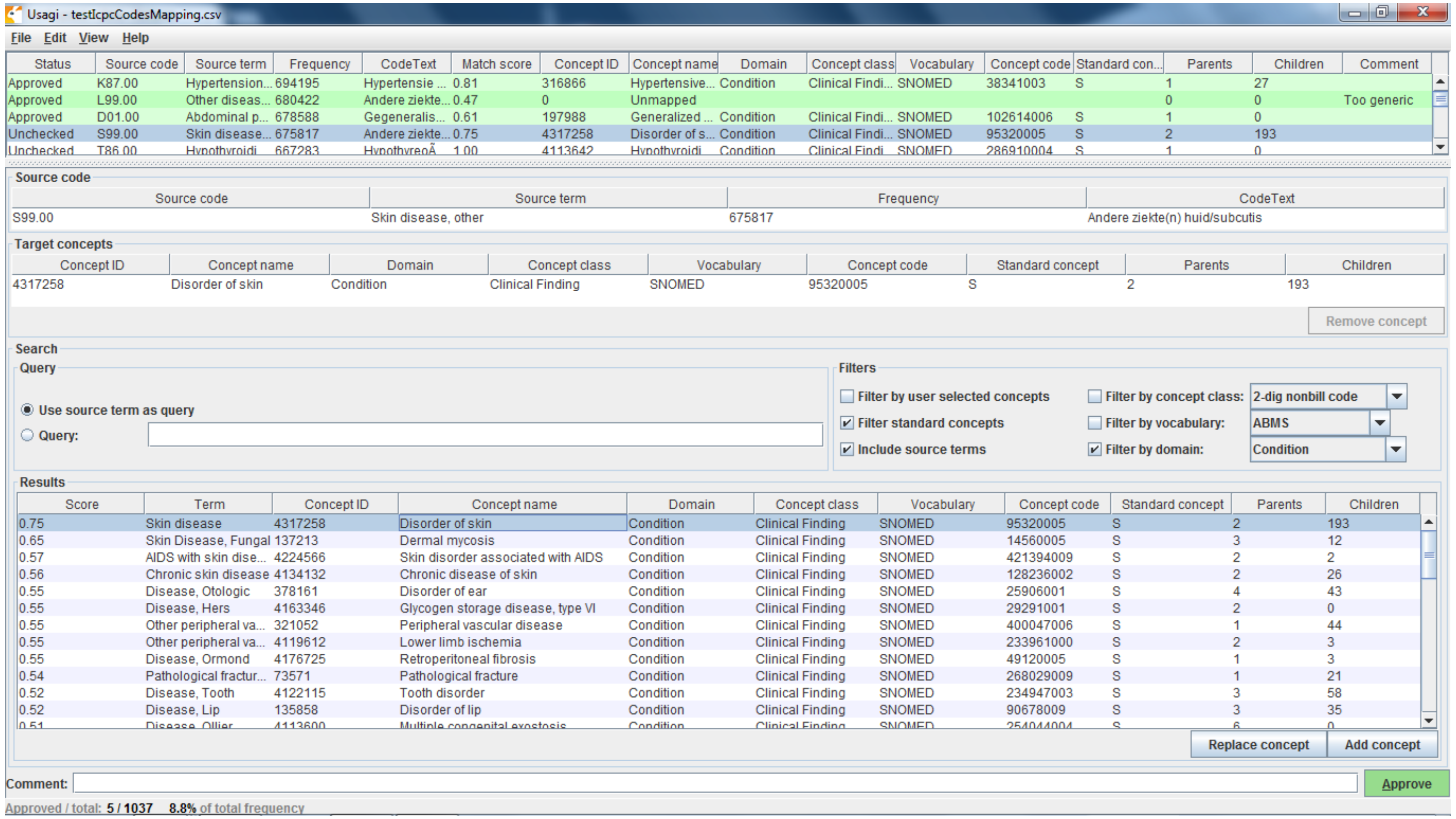
Rabbit -in-a-Hat
DATA MAPPING



https://github.com/OHDSI/WhiteRabbit

Usagi

CONCEPT MAPPING



https://github.com/OHDSI/Usagi

Usagi

Mapeo semántico a OMOP-CDM

USAGI
MAPEO SEMÁNTICO

Los códigos fuente que necesitan ser mapeados se cargan en Usagi.

Si los códigos no están en inglés, se requieren columnas adicionales con las traducciones.

Se utiliza un enfoque de similitud de términos para conectar los códigos fuente con los conceptos del Vocabulario.

Hay un paso previo de indexado de los vocabularios descargados de Athena

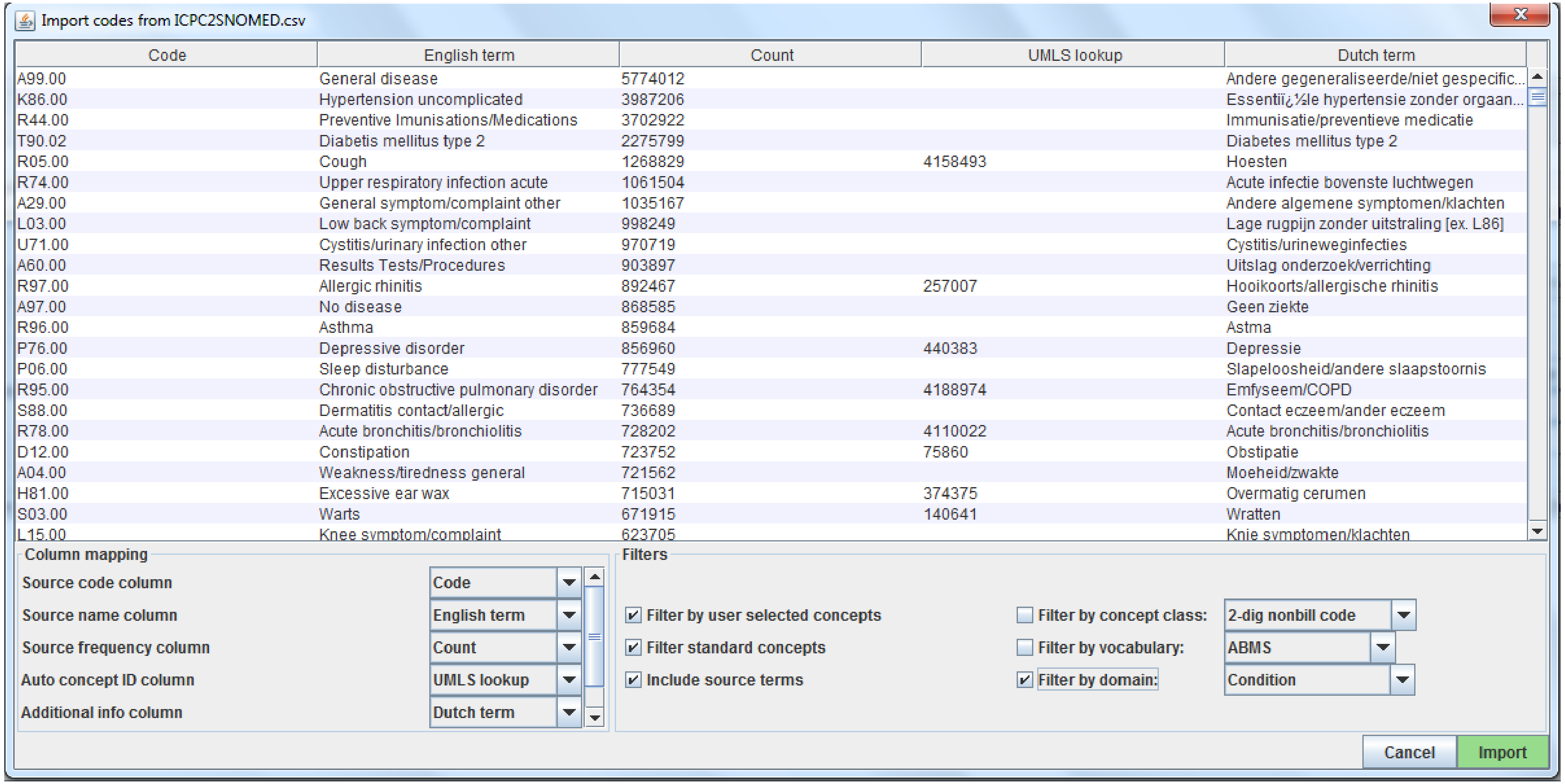
USAGI
MAPEO SEMÁNTICO

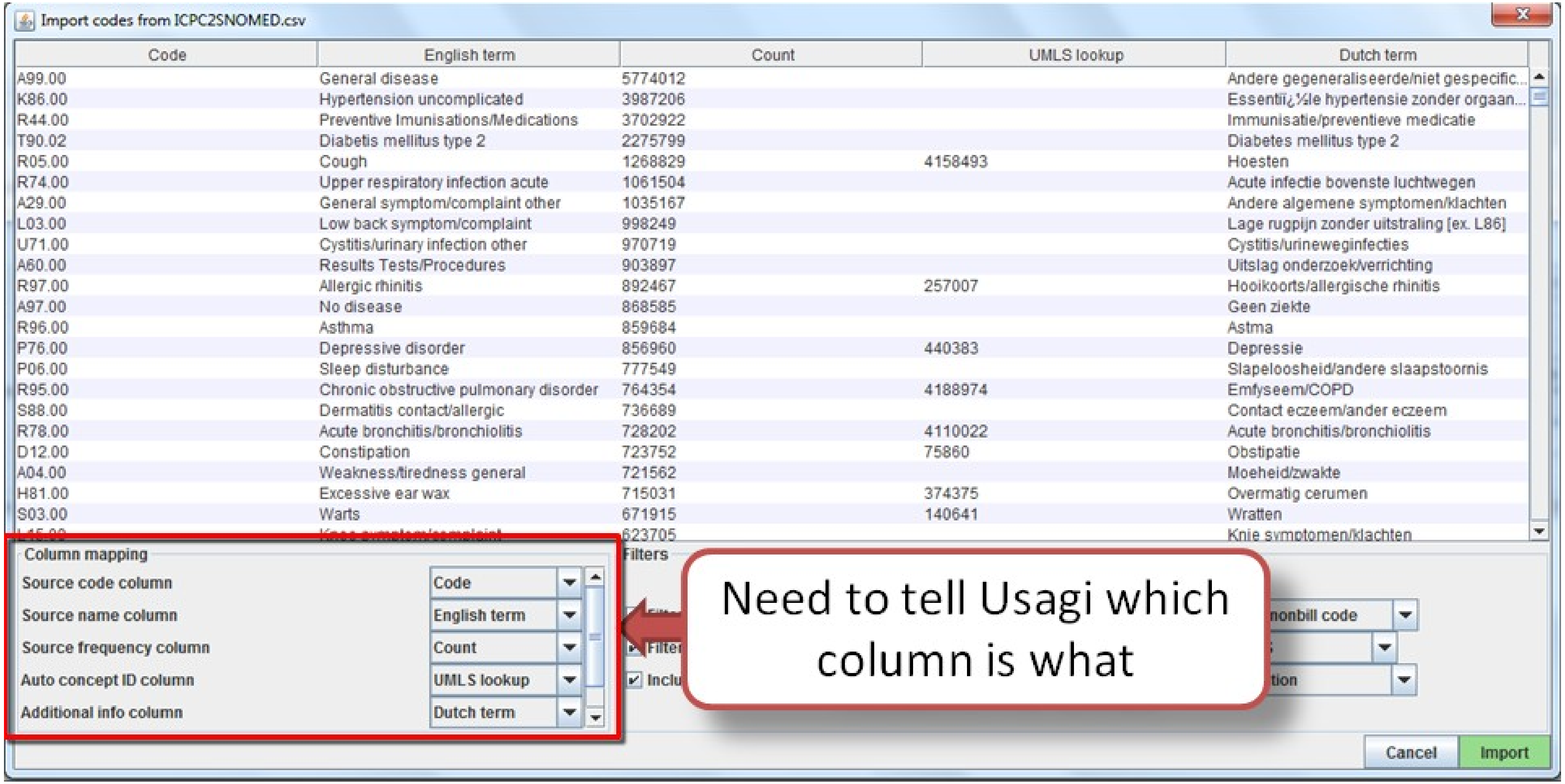
La secuencia típica para utilizar Usagi es:

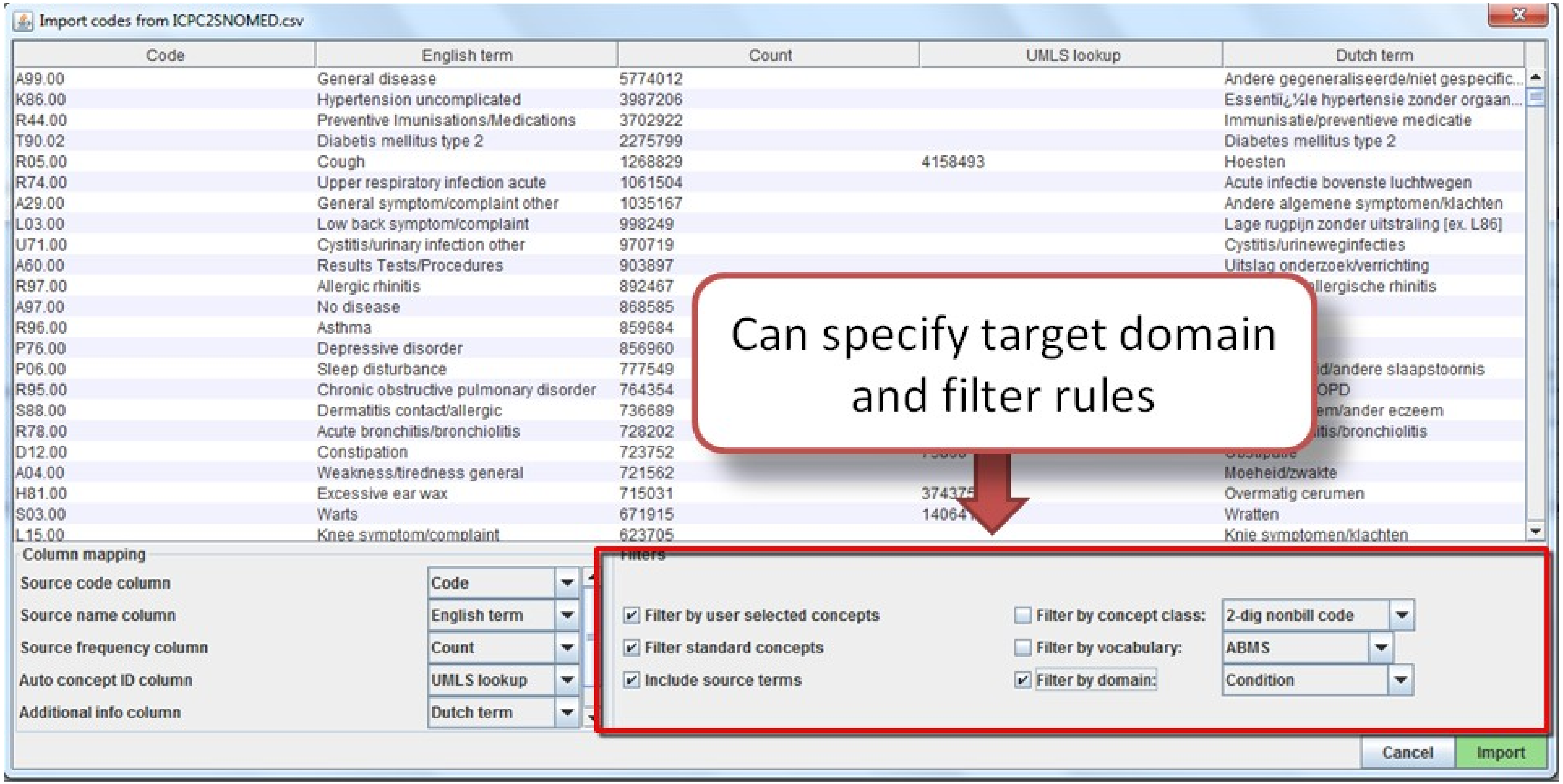
- Cargar los códigos fuente que deseamos mapear a conceptos del Vocabulario OMOP.

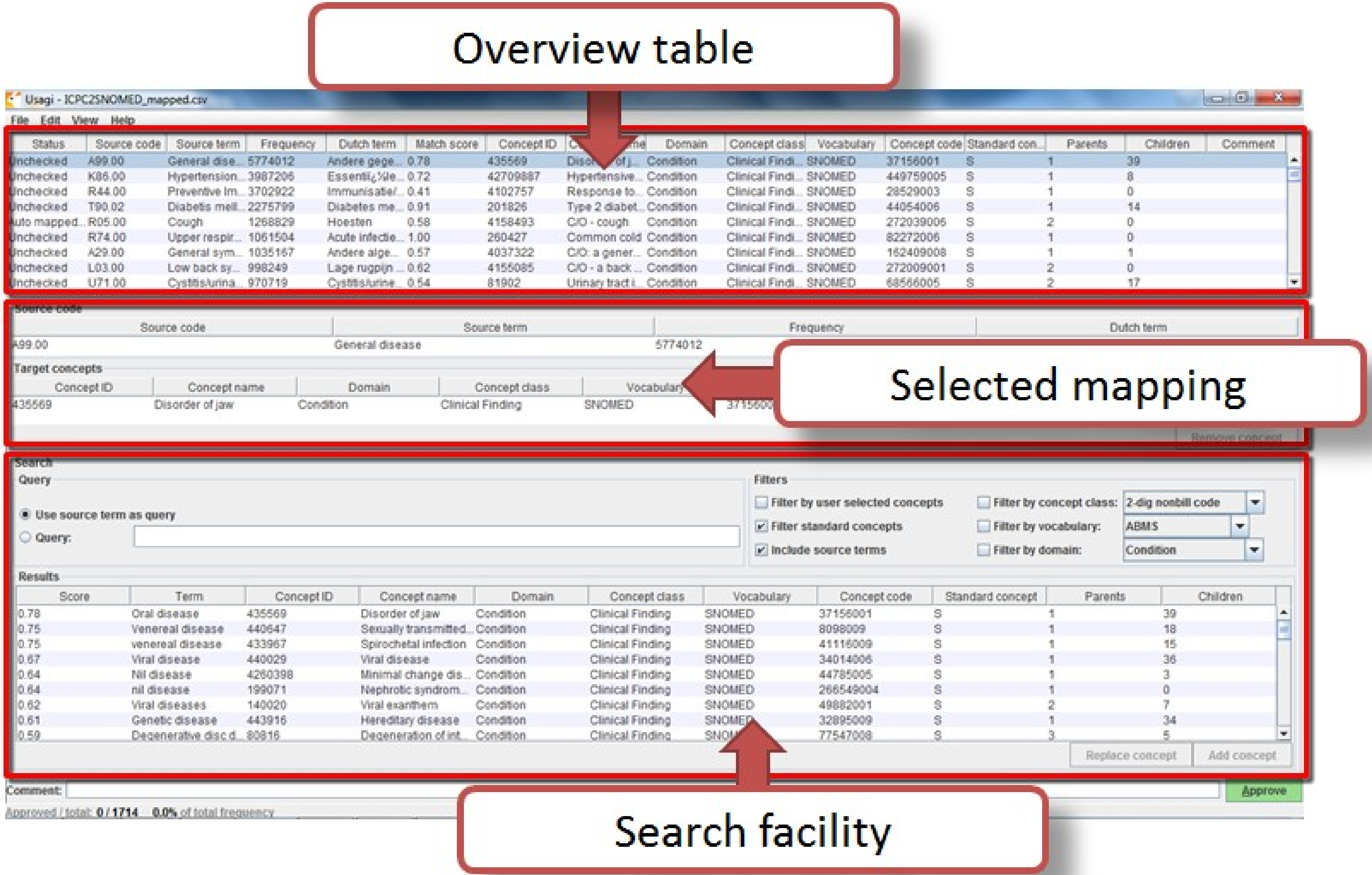
- Utilizaremos la interfaz de Usagi para revisar los mapeos sugeridos o crear nuevos mapeos. Generalmente, esta revisión debe realizarla una persona con experiencia en el sistema de codificación y en terminología médica.

- Exportar el mapeo final generado por Usagi a la tabla SOURCE_TO_CONCEPT_MAP del Vocabulario OMOP.


Import codes from ICPC2SNOMED.csv
Code | English term | Count | UMLS lookup | Dutch term
A99.00 | General disease | 5774012 | | Andere gegeneraliseerde/niet gespecific...
K86.00 | Hypertension uncomplicated | 3987206 | | Essentiï¿½le hypertensie zonder orgaan...
R44.00 | Preventive Imunisations/Medications | 3702922 | | Immunisatie/preventieve medicatie
T90.02 | Diabetis mellitus type 2 | 2275799 | | Diabetes mellitus type 2
R05.00 | Cough | 1268829 | 4158493 | Hoesten
R74.00 | Upper respiratory infection acute | 1061504 | | Acute infectie bovenste luchtwegen
A29.00 | General symptom/complaint other | 1035167 | | Andere algemene symptomen/klachten
L03.00 | Low back symptom/complaint | 998249 | | Lage rugpijn zonder uitstraling [ex. L86]
U71.00 | Cystitis/urinary infection other | 970719 | | Cystitis/urineweginfecties
A60.00 | Results Tests/Procedures | 903897 | | Uitslag onderzoek/verrichting
R97.00 | Allergic rhinitis | 892467 | 257007 | Hooikoorts/allergische rhinitis
A97.00 | No disease | 868585 | | Geen ziekte
R96.00 | Asthma | 859684 | | Astma
P76.00 | Depressive disorder | 856960 | 440383 | Depressie
P06.00 | Sleep disturbance | 777549 | | Slapeloosheid/andere slaapstoornis
R95.00 | Chronic obstructive pulmonary disorder | 764354 | 4188974 | Emfyseem/COPD
S88.00 | Dermatitis contact/allergic | 736689 | | Contact eczeem/ander eczeem
R78.00 | Acute bronchitis/bronchiolitis | 728202 | 4110022 | Acute bronchitis/bronchiolitis
D12.00 | Constipation | 723752 | 75860 | Obstipatie
A04.00 | Weakness/tiredness general | 721562 | | Moeheid/zwakte
H81.00 | Excessive ear wax | 715031 | 374375 | Overmatig cerumen
S03.00 | Warts | 671915 | 140641 | Wratten
L15.00 | Knee symptom/complaint | 623705 | | Knie symptomen/klachten
Column mapping
Source code column | Code
Source name column | English term
Source frequency column | Count
Auto concept ID column | UMLS lookup
Additional info column | Dutch term
Filters
Filter by user selected concepts
Filter standard concepts
Include source terms
Filter by concept class: | 2-dig nonbill code
Filter by vocabulary: | ABMS
Filter by domain: | Condition
Cancel
Import


Import codes from ICPC2SNOMED.csv
Code
English term
Count
UMLS lookup
Dutch term
A99.00 General disease 5774012 Andere gegeneraliseerde/niet gespecific...
K86.00 Hypertension uncomplicated 3987206 Essentiï¿½le hypertensie zonder orgaan...
R44.00 Preventive Imunisations/Medications 3702922 Immunisatie/preventieve medicatie
T90.02 Diabetis mellitus type 2 2275799 Diabetes mellitus type 2
R05.00 Cough 1268829 4158493 Hoesten
R74.00 Upper respiratory infection acute 1061504 Acute infectie bovenste luchtwegen
A29.00 General symptom/complaint other 1035167 Andere algemene symptomen/klachten
L03.00 Low back symptom/complaint 998249 Lage rugpijn zonder uitstraling [ex. L86]
U71.00 Cystitis/urinary infection other 970719 Cystitis/urineweginfecties
A60.00 Results Tests/Procedures 903897 Uitslag onderzoek/verrichting
R97.00 Allergic rhinitis 892467 257007 Hooikoorts/allergische rhinitis
A97.00 No disease 868585 Geen ziekte
R96.00 Asthma 859684 Astma
P76.00 Depressive disorder 856960 440383 Depressie
P06.00 Sleep disturbance 777549 Slapeloosheid/andere slaapstoornis
R95.00 Chronic obstructive pulmonary disorder 764354 4188974 Emfyseem/COPD
S88.00 Dermatitis contact/allergic 736689 Contact eczeem/ander eczeem
R78.00 Acute bronchitis/bronchiolitis 728202 4110022 Acute bronchitis/bronchiolitis
D12.00 Constipation 723752 75860 Obstipatie
A04.00 Weakness/tiredness general 721562 Moeheid/zwakte
H81.00 Excessive ear wax 715031 374375 Overmatig cerumen
S03.00 Warts 671915 140641 Wratten
623705 Knie symptomen/klachten
Column mapping
Source code column Code
Source name column English term
Source frequency column Count
Auto concept ID column UMLS lookup
Additional info column Dutch term
Filters
Need to tell Usagi which column is what
Cancel
Import


Import codes from ICPC2SNOMED.csv
Code
English term
Count
UMLS lookup
Dutch term
A99.00 General disease 5774012 Andere gegeneraliseerde/niet gespecific...
K86.00 Hypertension uncomplicated 3987206 Essentiï¿½le hypertensie zonder orgaan...
R44.00 Preventive Imunisations/Medications 3702922 Immunisatie/preventieve medicatie
T90.02 Diabetis mellitus type 2 2275799 Diabetes mellitus type 2
R05.00 Cough 1268829 4158493 Hoesten
R74.00 Upper respiratory infection acute 1061504 Acute infectie bovenste luchtwegen
A29.00 General symptom/complaint other 1035167 Andere algemene symptomen/klachten
L03.00 Low back symptom/complaint 998249 Lage rugpijn zonder uitstraling [ex. L86]
U71.00 Cystitis/urinary infection other 970719 Cystitis/urineweginfecties
A60.00 Results Tests/Procedures 903897 Uitslag onderzoek/verrichting
R97.00 Allergic rhinitis 892467
A97.00 No disease 868585
R96.00 Asthma 859684
P76.00 Depressive disorder 856960
P06.00 Sleep disturbance 777549
R95.00 Chronic obstructive pulmonary disorder 764354
S88.00 Dermatitis contact/allergic 736689
R78.00 Acute bronchitis/bronchiolitis 728202
D12.00 Constipation 723752
A04.00 Weakness/tiredness general 721562 Moeheid/zwakte
H81.00 Excessive ear wax 715031 Overmatig cerumen
S03.00 Warts 671915 Wratten
L15.00 Knee symptom/complaint 623705 Knie symptomen/klachten
Can specify target domain and filter rules
Column mapping
Source code column Code
Source name column English term
Source frequency column Count
Auto concept ID column UMLS lookup
Additional info column Dutch term
Filter by user selected concepts
Filter standard concepts
Include source terms
Filter by concept class: 2-dig nonbill code
Filter by vocabulary: ABMS
Filter by domain: Condition
Cancel
Import



Overview table
Usagi - ICPC2SNOMED_mapped.csv
File Edit View Help
Source code
Target concepts
Selected mapping
Search
Query
Use source term as query
Query:
Filters
Filter by user selected concepts
Filter standard concepts
Include source terms
Filter by concept class:
Filter by vocabulary:
Filter by domain:
Results
Replace concept
Add concept
Comment:
Approve
Search facility

DBT

Transformación a OMOP-CDM

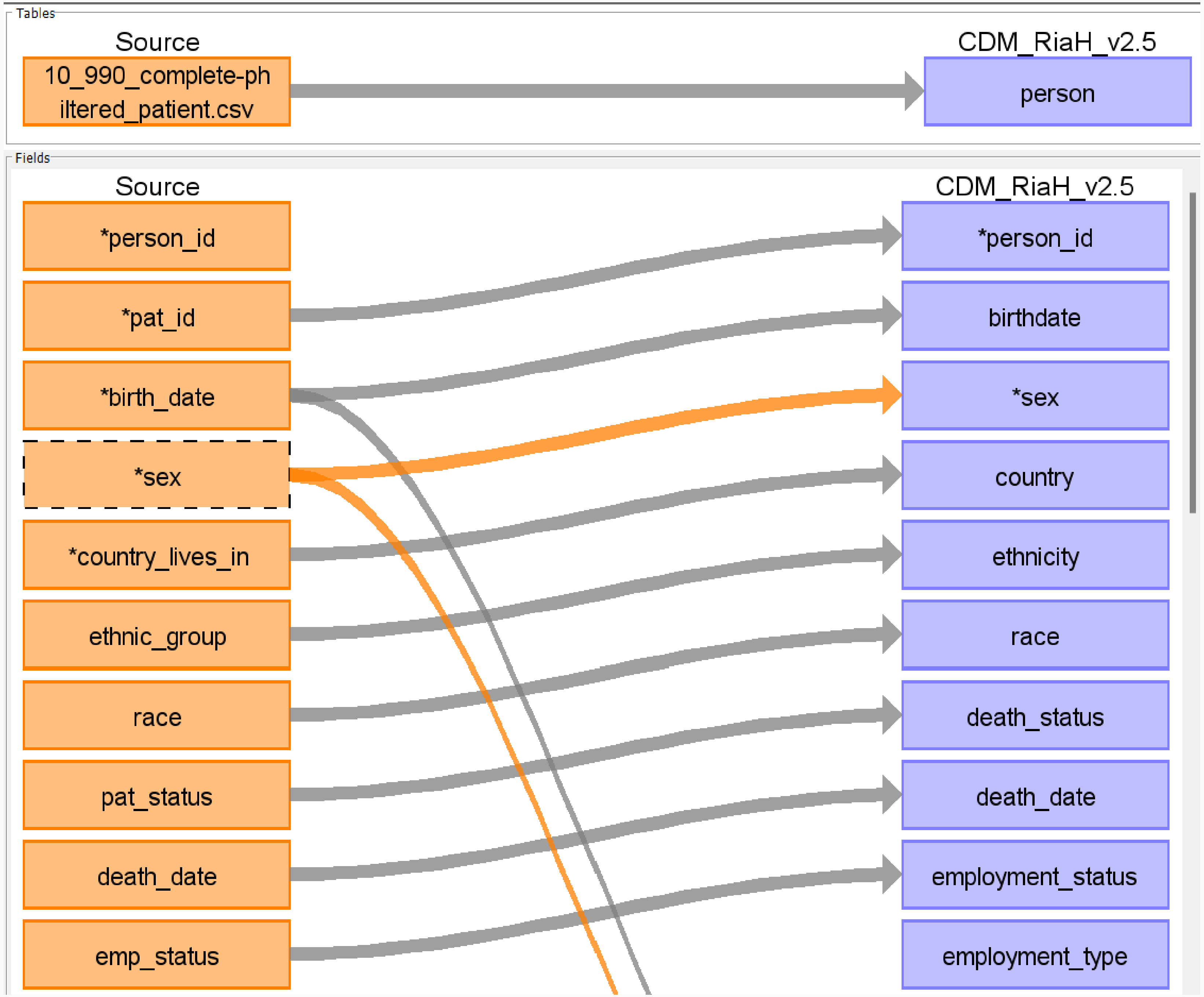
Sources

```yaml
version: 2

sources:
  - name: raw
    schema: raw
    tables:
      - name: 0_990_Complete-philtered_COMP_LABS_OLIGOCLONAL
      - name: 1_990_Complete-philtered_CONFIRMED BY
      - name: 2_990_Complete-philtered_Classification
      - name: 3_990_Complete-philtered_Clinical_Attacks
      - name: 990_Complete-philtered_H_W
      - name: 7_990_Complete-philtered_MRI_ACTIVITY
      - name: 990_Complete-philtered_OCB
      - name: 9_990_Complete-philtered_Order_med
      - name: 10_990_Complete-philtered_Patient
      - name: 11_990_Complete-philtered_Physical Exam Metrics
      - name: 12_990_Complete-philtered_RxNorm
      - name: 13_990_Complete-philtered_SDE_DMT
      - name: 14_990_Complete-philtered_Smoking
      - name: 15_990_Complete-philtered_Symptoms
      - name: 16_990_Complete-philtered_row19-Diagnosis Date

    #   - name: 6_990_Complete-philtered_Interval History
    #   - name: 4_990_Complete-philtered_ENCOUNTERS
```

Models

```
{{ config(
    materialized='table'
) }}

select
"PERSON_ID",
"PAT_ID" pat_id,
"LINE",
"SFGROUP",
"MEDICATIONS" medication,
"MED START DATE"::DATE med_start_date,
"MED END DATE"::DATE med_start_date,
"MED DISCONTINUATION REASON" med_discontinuation_reason
from {{ source('raw', '13_990_Complete-philtered_SDE_DMT') }}
```

Models


Tables
Source
10_990_complete-philtered_patient.csv
CDM_RiaH_v2.5
person
Fields
Source
*person_id
*pat_id
*birth_date
*sex
*country_lives_in
ethnic_group
race
pat_status
death_date
emp_status
CDM_RiaH_v2.5
*person_id
birthdate
*sex
country
ethnicity
race
death_status
death_date
employment_status
employment_type

```
with patient as (
    select *
    from {{ ref('raw_patient') }}
),
```

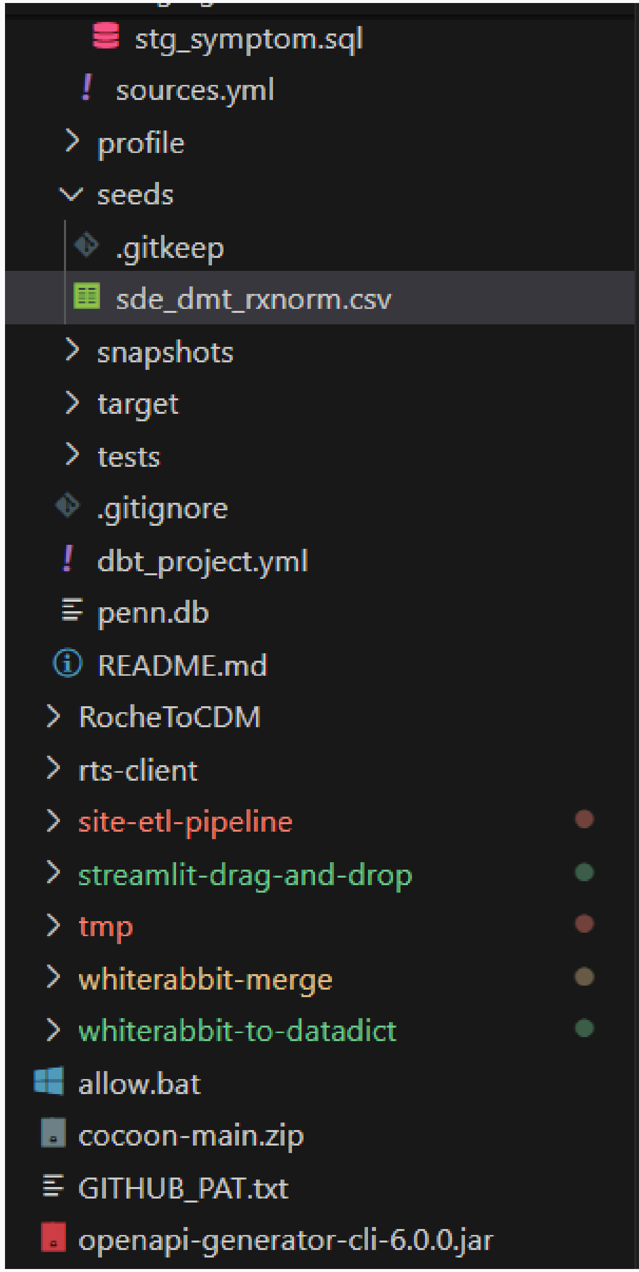
```
SELECT
    patient."PAT_ID" AS person_id,
    patient."BIRTH_DATE" AS birthdate,
    {{ sex_mapping('patient."SEX"') }} AS sex,
    {{ country_mapping('patient."COUNTRY_LIVES_IN"') }} AS country,
    {{ ethnicity_mapping('patient."ETHNIC_GROUP"') }} AS ethnicity,
    {{ race_mapping('patient."RACE"') }} AS race,
    {{ death_status_mapping('patient."PAT_STATUS"') }} AS death_status,
    patient."DEATH_DATE" AS death_date,
    {{ employment_status_mapping('patient."EMP_STATUS"') }} AS employment_status,
    {{ employment_type_mapping('patient."EMP_STATUS"') }} AS employment_type,
    NULL::TEXT AS inactivity_reason,
    NULL::TEXT AS education_level,
    {{ smoking_status_mapping('patient."SMOKING_STATUS"') }} AS smoking_status,
    patient."PAT_ID" AS person_id_raw_value,
    patient."BIRTH_DATE" AS birthdate_raw_value,
    patient."SEX" AS sex_raw_value,
    patient."COUNTRY_LIVES_IN" AS country_raw_value,
    patient."ETHNIC_GROUP" AS ethnicity_raw_value,
    patient."RACE" AS race_raw_value,
    patient."PAT_STATUS" AS death_status_raw_value,
    patient."DEATH_DATE" AS death_date_raw_value,
    patient."EMP_STATUS" AS employment_status_raw_value,
    patient."EMP_STATUS" AS employment_type_raw_value,
    NULL::TEXT AS inactivity_reason_raw_value,
    NULL::TEXT AS education_level_raw_value,
    smoking."SMOKING_STATUS" AS smoking_status_raw_value
FROM patient
```

Models

Macros

```
--- Macro to transform 'H' and 'M' sex values into their concept_id
{% macro sex_mapping(sex) %}
(CASE WHEN {{ sex }} = 'MALE' THEN 'Male' -- man
      WHEN {{ sex }} = 'FEMALE' THEN 'Female' -- woman
      ELSE NULL::TEXT -- unknown
      END)
{% endmacro %}
```

```
with rxnorm as (
    select medication,rxnorm_code
    from {{ ref('sde_dmt_rxnorm') }}
)
```

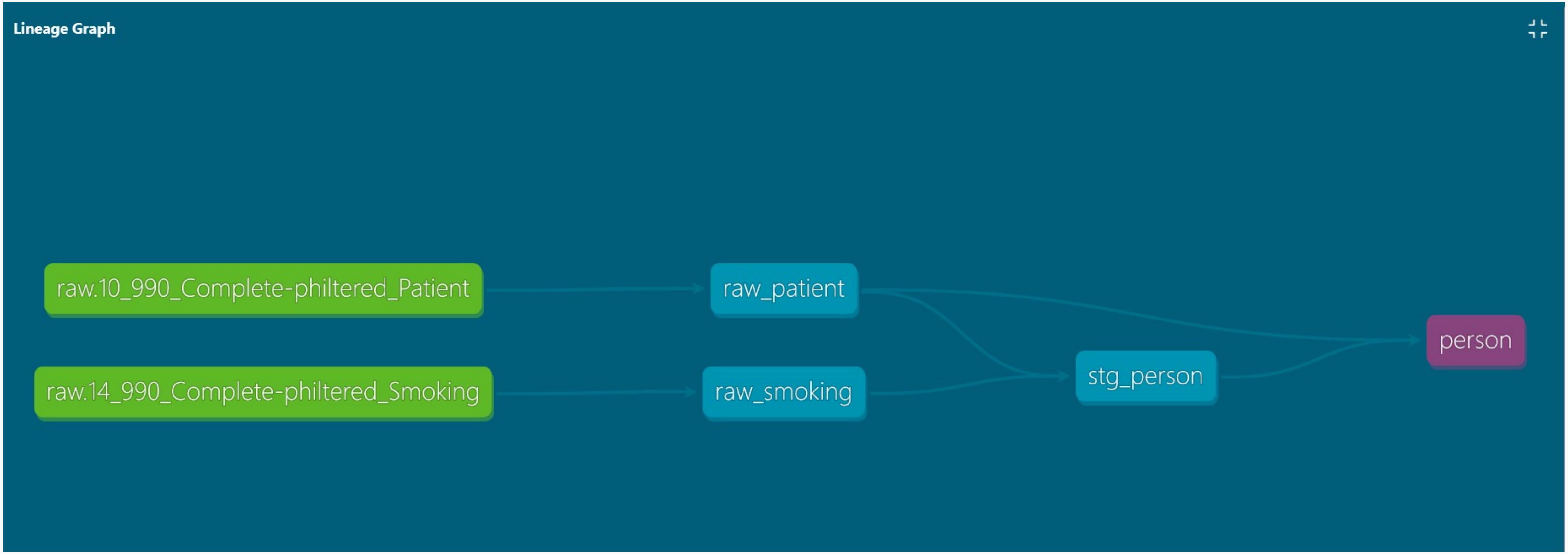


	column 1	column 2
1	medication	rxnorm_code
2	Ocrevus	1876366
3	Copaxone (40mg)	214582
4	Tecfidera (240mg)	1373490
5	Kesimpta	712566
6	Avonex	75917
7	Copaxone (20mg)	214582
8	Gilenya	1012892
9	Tysabri	354770
10	Aubagio	1310520
11	Betaferon	72257
12	Rebif (44)	75917
13	Mavenclad	44157
14	Rituxan (every 6 months)	121191
15	Zeposia	2288236
16	Vumerity	2261783
17	Mayzent	2121085
18	Tecfidera (120mg)	1373479
19	Plegridy	1546168

Seeds

Profile

```yaml
penn:
  target: dev
  outputs:
    penn:
      type: postgres
      threads: 16
      host: "{{ env_var('HOST') }}"
      port: "{{ env_var('PORT') }}"
      user: "{{ env_var('USER') }}"
      pass: "{{ env_var('PASS') }}"
      dbname: "{{ env_var('DBNAME') }}"
      schema: "{{ env_var('SCHEMA') }}"

    dev:
      type: sqlite
      threads: 1
      database: 'penn.db'
      schema: cdm
      schemas_and_paths:
        main: 'penn.db'
      schema_directory: 'models'
```



Documentation

Command	Description
dbt seed	Loads data from CSV files into your data warehouse. These CSV files are typically located in the data directory and are helpful for small, static datasets.
dbt compile	Compiles the dbt models into SQL files without running them against the database. Useful for checking that models are correctly set up.
dbt run	Executes the compiled models in the data warehouse, transforming raw data according to the specified models.
dbt docs	Generates documentation for the dbt project, including details about models, lineage, and dependencies.

Extras

Transformación a OMOP-CDM

PROFILING

ydata-profiling

patients Profiling Report

Overview Variables Interactions Correlations Missing values Sample

Overview

Brought to you by YData

Overview Alerts 21 Reproduction

Dataset statistics

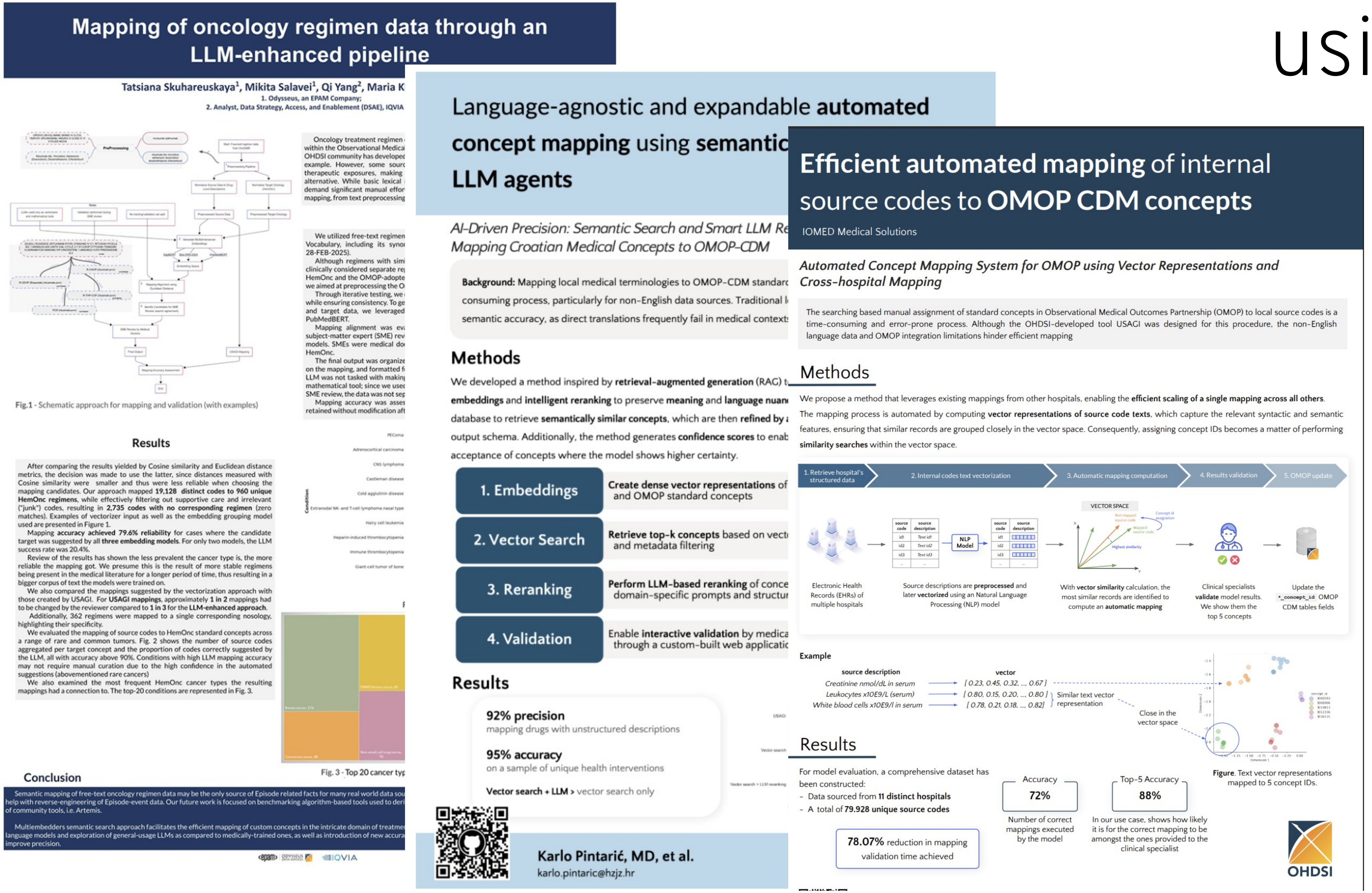
Number of variables	28
Number of observations	111
Missing cells	454
Missing cells (%)	14.6%
Duplicate rows	0
Duplicate rows (%)	0.0%
Total size in memory	24.4 KiB
Average record size in memory	225.2 B

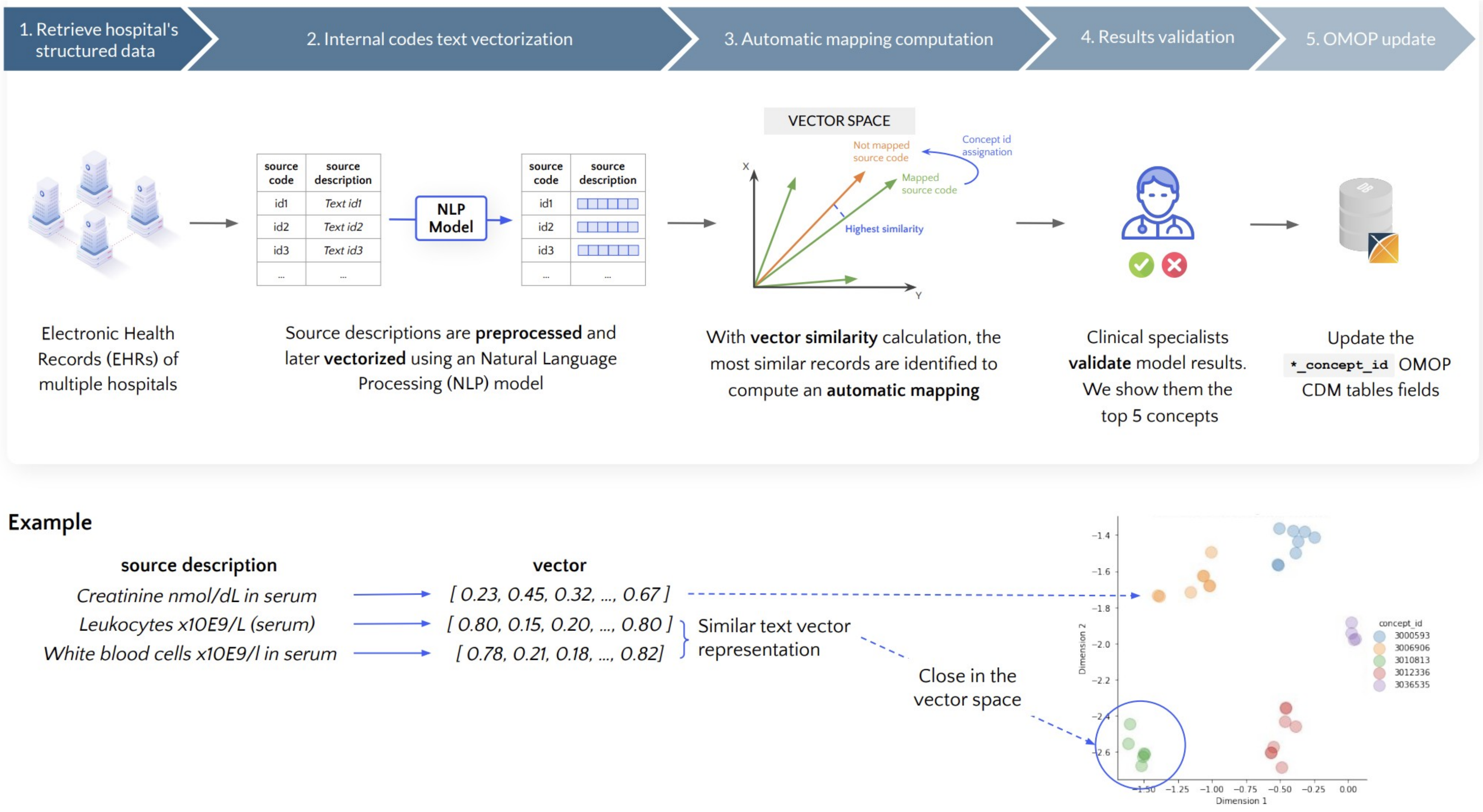
Variable types

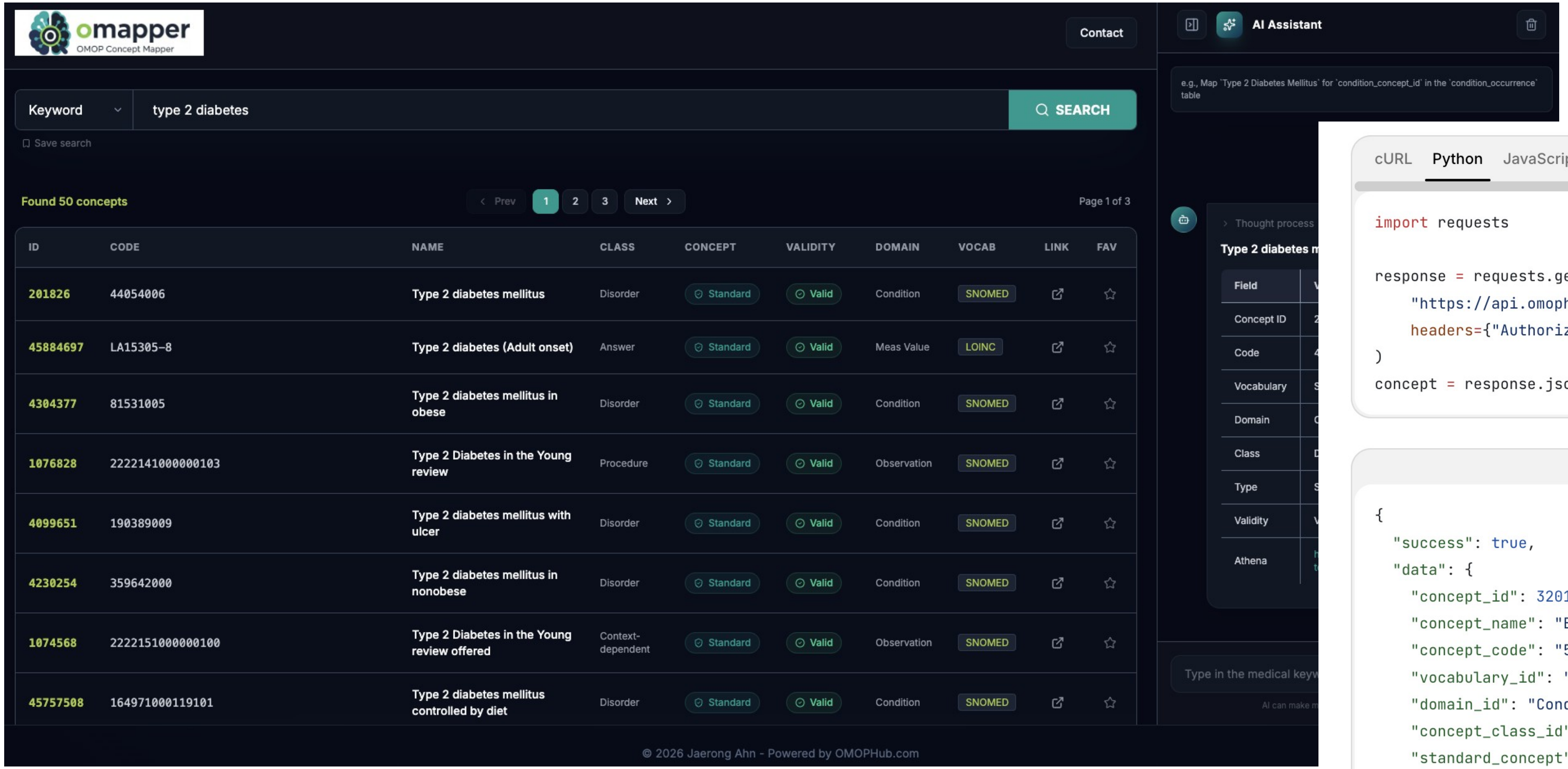
Text	21
Numeric	7

https://docs.profiling.ydata.ai/latest/

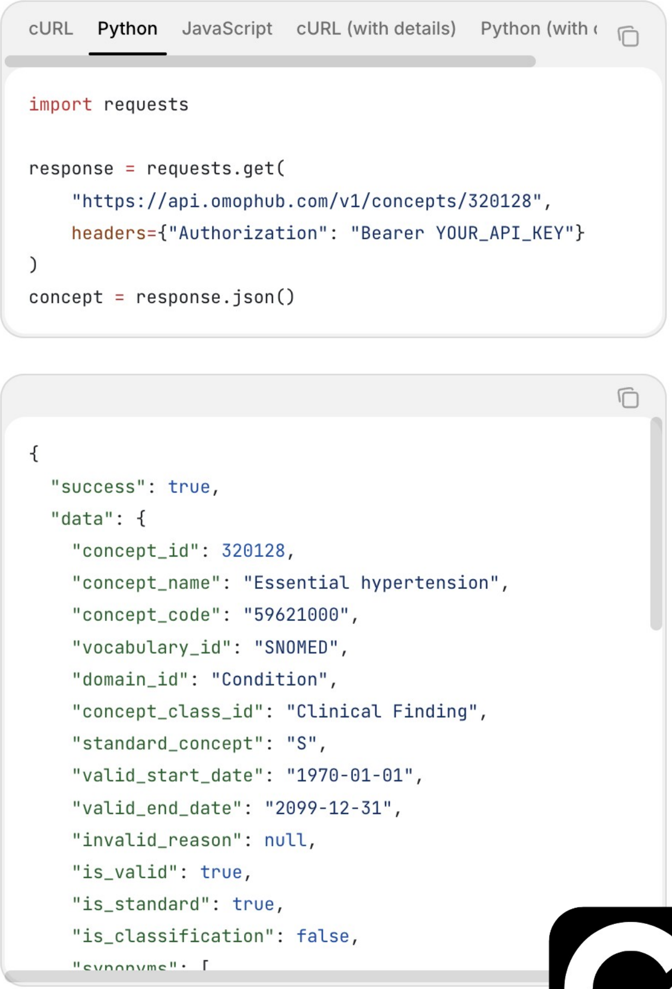
MAPPING
using LLMs



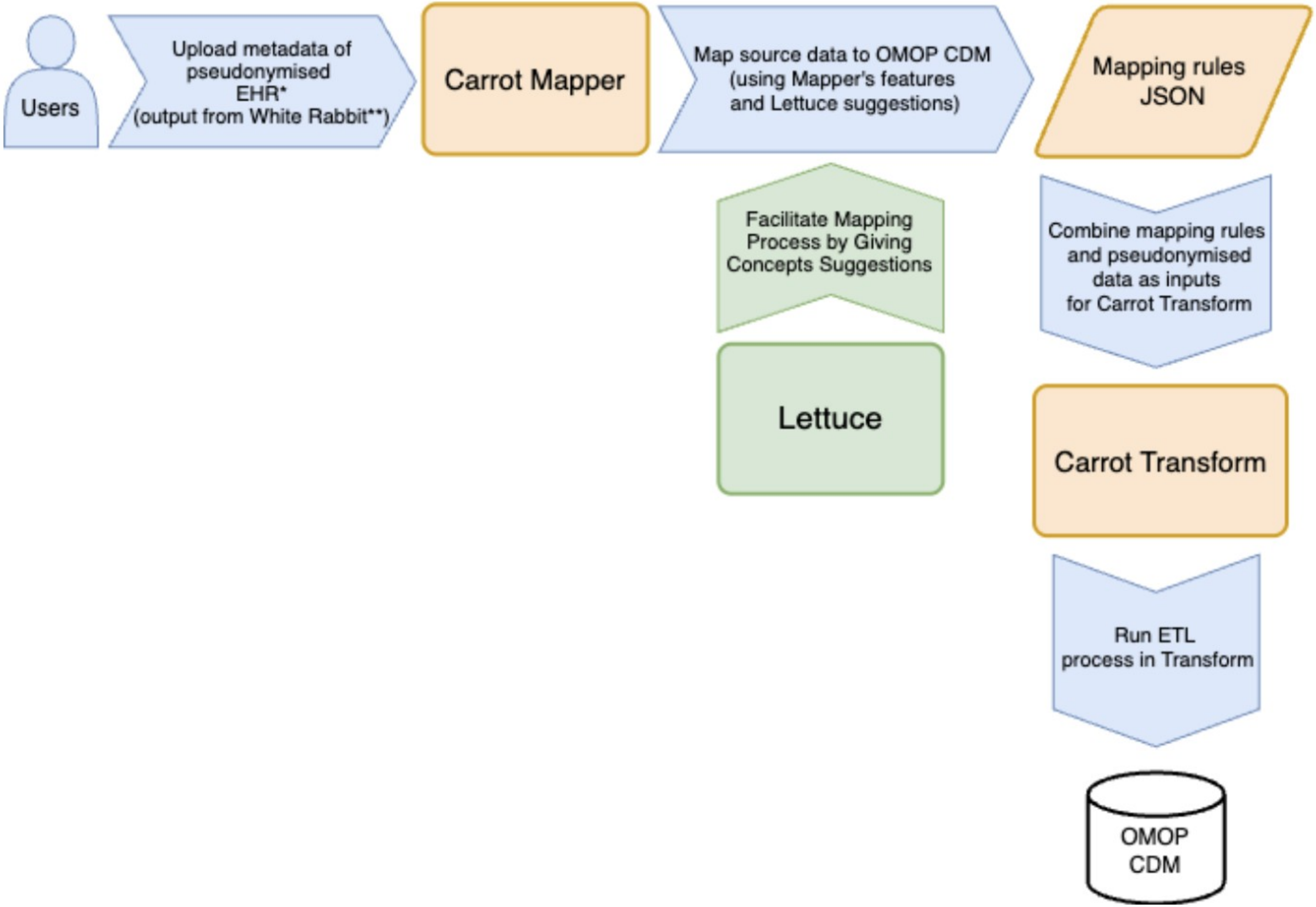

1. Retrieve hospital's structured data
2. Internal codes text vectorization
3. Automatic mapping computation
4. Results validation
5. OMOP update
source code
source description
id1
Text id1
id2
Text id2
id3
Text id3
NLP Model
VECTOR SPACE
Not mapped source code
Concept id assignation
Mapped source code
Highest similarity
Electronic Health Records (EHRs) of multiple hospitals
Source descriptions are **preprocessed** and later **vectorized** using an Natural Language Processing (NLP) model
With **vector similarity** calculation, the most similar records are identified to compute an **automatic mapping**
Clinical specialists **validate** model results. We show them the top 5 concepts
Update the `*_concept_id` OMOP CDM tables fields
Example
source description
vector
Creatinine nmol/dL in serum
Leukocytes x10E9/L (serum)
White blood cells x10E9/l in serum
[0.23, 0.45, 0.32, ..., 0.67]
[0.80, 0.15, 0.20, ..., 0.80]
[0.78, 0.21, 0.18, ..., 0.82]
Similar text vector representation
Close in the vector space
Dimension 1
Dimension 2
concept_id
3000593
3006906
3010813
3012336
3036535



https://www.omapper.org/



https://omophub.com/



https://github.com/Health-Informatics-UoN

Preguntas?

GRACIAS POR VUESTRA ATENCIÓN